A project report on

DEEPFAKE MODEL ATTRIBUTION VIA GAN ENCODER LATENT SIGNATURES FOR FACES

Submitted in partial fulfillment for the award of the degree of

Master of Computer Applications

by

HAMI YASIR (24MCA1020)



VIT®
Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)
CHENNAI

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

April, 2026

DEEPFAKE MODEL ATTRIBUTION VIA GAN ENCODER LATENT SIGNATURES FOR FACES

Submitted in partial fulfillment for the award of the degree of

Master of Computer Applications

by

HAMI YASIR (24MCA1020)



VIT®

Vellore Institute of Technology

(Deemed to be University under section 3 of UGC Act, 1956)

CHENNAI

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

April, 2026



VIT®

Vellore Institute of Technology

(Deemed to be University under section 3 of UGC Act, 1956)

CHENNAI

School of Computer Science and Engineering

DECLARATION

I hereby declare that the thesis entitled "DEEPFAKE MODEL ATTRIBUTION VIA GAN ENCODER LATENT SIGNATURES FOR FACES" submitted by me, for the award of the degree of Master of Computer Applications, Vellore Institute of Technology ,Chennai is a record of bonafide work carried out by me under the supervision of Guide Name

I further declare that the work reported in this thesis has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university.

Place: Chennai

Date:

Signature of the Candidate

VIT®

Vellore Institute of Technology

(Deemed to be University under section 3 of UGC Act, 1956)

CHENNAI

School of Computer Science and Engineering

CERTIFICATE

This is to certify that the report entitled **"**DEEPFAKE MODEL ATTRIBUTION VIA GAN ENCODER LATENT SIGNATURES FOR FACES**"** is prepared and submitted by **Hami Yasir** (**24MCA1020**) to VIT Chennai, in partial fulfillment of the requirement for the award of the degree of **Master of Computer Applications,** programme and as a part of PMCA699J - Dissertation II is a bonafide record carried out under my guidance. The project fulfills the requirements as per the regulations of this University and in my opinion meets the necessary standards for submission. The contents of this report have not been submitted and will not be submitted either in part or in full, for the award of any other degree or diploma and the same is certified.

Signature of the Guide

Name: Dr. Rishikeshan C A

Date:

Signature of the Internal Examiner

Name:

Date:

Signature of the External Examiner

Name:

Date:

Approved by the Head of Department,
Master of Computer Applications

Name: **Dr. Graceline Jasmine S**
Associate Professor

Date:

(Seal of SCOPE)

ABSTRACT

As a result of rapid advances in Generative Adversarial Networks (GAN's), the amount of realistic synthetic images has increased substantially. This has raised significant concerns about the potential for misinformation, misuse of identity, and digital forensics. Most of the research currently being conducted on GAN's has focused on developing methods for determining if a given image came from a real or fake source. However, comparatively little attention has been paid to developing methods for determining the specific type of generator that produced an image. This work describes a new method for helping fill this gap through the use of a GAN Attribution Framework Based on Latent-Space Forensics. This framework uses an encoder-based inversion technique (e4e) to map generated images into the W+ latent space while capturing layer wise generator-specific signatures through statistical analysis of the images. These latent fingerprints are then used to construct explainable machine learning models to allow researchers to attribute the generated images back to the original GAN architecture that generated them. Through empirical testing, we will demonstrate the effectiveness of latent-space statistical signatures as a reliable source of distinctive features for GAN architecture attributions, making for a clearer and more scalable solution to attributing GAN generated images compared with traditional forensic methods that rely on analyzing the images themselves.

ACKNOWLEDGEMENT

It is my pleasure to express with a deep sense of gratitude to Dr. Rishikeshan C A, Associate Professor, SCOPE, Vellore Institute of Technology, Chennai, for his constant guidance, continual encouragement, and understanding; more than all, he taught me patience in my endeavor. My association with him is not confined to academics only, but it is a great opportunity on my part of work with an intellectual and expert in the field of Machine Learning.

It is with gratitude that I would like to extend thanks to our honorable Chancellor, Dr. G. Viswanathan, Vice Presidents, Mr. Sankar Viswanathan, Dr. Sekar Viswanathan and Dr. G V Selvam, Executive Director Dr. Sandhya Pentareddy, Assistant Vice-President, Ms. Kadhambari S. Viswanathan, Vice-Chancellor, Dr. V. S. Kanchana Bhaaskaran, Pro-Vice Chancellor Dr.T. Thyagarajan and Additional Registrar, Dr. P.K.Manoharan for providing an exceptional working environment and inspiring all of us during the tenure of the course.

Special mention to Dean, Dr.V.Viswanathan, Associate Dean, Dr.P.Nithyanandam, Associate Dean, Dr.G.Suganya, Associate Dean, Dr.C.Sweetlin Hemalatha, SCOPE, Vellore Institute of Technology, Chennai, for spending their valuable time and efforts in sharing their knowledge and for helping us in every aspect.

In a jubilant mood, I express ingeniously my whole-hearted thanks to Dr.S.Graceline Jasmine, Head of the Department and Dr.R.Kanniga Devi, Project Coordinator, SCOPE, Vellore Institute of Technology, Chennai, for their valuable support and encouragement to take up and complete the thesis.

My sincere thanks to all the faculty member and staffs at Vellore Institute of Technology, Chennai, who helped me acquire the requisite knowledge. I would like to thank my parents for their support. It is indeed a pleasure to thank my friends who encouraged me to take up and complete this task.

Place: Chennai

Date: **Hami Yasir**

CONTENTS Page No:

CONTENTS..... iv

LIST OF FIGURES ix

LIST OF TABLES xi

LIST OF ACRONYMS xii

CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION 1

1.2 OVERVIEW ……..... 3

1.3 CHALLENGES……..... 4

1.4 PROJECT STATEMENT..... 6

1.5 OBJECTIVES 7

1.6 SCOPE OF THE PROJECT 8

CHAPTER 2

BACK GROUND

2.1 LITERATURE SURVEY..... 9

2.2 RESEARCH TRENDS..... 39

CHAPTER 3

PROJECT

3.1 PROJECT DOMAIN…..... 43

3.2 DATASET……..... 44

3.3 GENERATIVE ADVERSARIAL NETWORKS (GANs) …….....45

3.4 HARDWARE AND SOFTWARE REQUIREMENTS …….....47

3.5 METHODOLOGY …….....50

CHAPTER 4

CONCLUSION & FUTURE WORK

4.1 CONCLUSION…..…..... 64

4.2 FUTURE WORKS.…..... 65

LIST OF FIGURES

1.1 METHODOLOGY PIPELINE..... 52

1.2 ARCHITECTURE DIAGRAM..... 53

1.3 LINEAR REGRESSION CONFUSION MATRIX..... 55

1.4 LINEAR SVM CONFUSION MATRIX..... 56

1.5 RANDOM FOREST CONFUSION MATRIX..... 57

LIST OF TABLES

2.1 LOGISTIC REGRESSION CLASSIFICATION REPORT..... 54

2.2 LINEAR SVM CLASSIFICATION REPORT..... 56

2.3 RANDOM FOREST CLASSIFICATION REPORT..... 57

LIST OF ACRONYMS

GANs	Generative Adversarial Networks
SVM	Support Vector Machine
CNN	Convolutional Neural Networks
DCT	Discrete Cosine Transform
PRNUs	Photo Response Non-Uniformity
CLIP	Contrastive Language Image Pretraining
DFT	Discrete Fourier Transform
FFHQ	Flickr Faces High Quality Dataset
AI	Artificial Intelligence

Chapter 1

Introduction

1.1 INTRODUTION

The rapid expansion of generative adversarial networks (GANs) has led to the mass availability of photo, realistic synthetic pictures from different digital media sources. Although these technological innovations can be used to create many works and bring a lot of benefits to different industries, they raise quite a few problems for digital forensics, the spreading of false news, and checking the authenticity of content. It is a crucial matter to be able to track back the place from which a generated photo came to be in order to be able to make companies take responsibility for their actions, thus providing a better system for the protection of intellectual property rights, and in this way, one of relying on the visual media can be established. Nevertheless, most of the existing methods of analysis are directed at whether an image is real or synthetic, not to the extent of which generative model it was that created it; thus, leaving a significant void that can be filled with further research in digital forensics.

The objective of this project is to tackle GAN source attribution issues using latent-space forensics as a solution. Instead of using image-space artefacts, the solutions proposed in this project use generator-intrinsic signatures contained in latent representations. The solutions presented in this project use an encoder-based inversion method whereby generated images are transformed into W+ latent spaces, thus allowing for extracting of the model-specific statistical signature characteristics that represent the behaviour of specific models. The latent representations provide an ordered and interpretable basis from which to learn the discriminative fingerprints assigned to each GAN architecture.

A new methodology that utilizes a set of machine-learning-based algorithms trained via latent-space statistics extracted from encoder-projected representations. Logistic

Regression and Random Forests, both explainable classifiers, have been built to distinguish among various architectures at the generative level using architecture-specific and layer-specific statistics, as well as per-image latent values. This architecture provides robust predictions while minimising reliance on non-robust artefacts present in images and allows transparent evaluation of the basis on which these predictions are made. The result is a model that, after training the machine learning algorithms, will allow users to submit an image generated from one of the model's generator architectures, whereby the user will receive an output specifying what the underlying GAN generator architecture was most likely to be, as well as an attributed confidence score.

The principal challenge associated with the use of this method is in getting the correct accuracy rate and correct representations of the latent features of high-dimensional data in an efficient manner. Inversion must produce the output that correctly represents and captures the features of the underlying generator, with the least error in terms of accuracy. Feature engineering and model validation will be vital to resolving these challenges owing to the immense size of the latent dataset and need for robustness across different architecture types and training scenarios. Without solving these challenges, it will not be possible to build an attribution system that can be scaled and generalised.

This study proposes a method for attributing the source of an image generated by several different GAN models by creating a latent-space signature for each image. The proposed method will utilize the W+ space to generate the latent representations of an input image, and then use the generator layer statistics to determine the GAN model that generated the input image. This will provide a practical and interpretable means of determining the source of an image generated by a GAN for use in forensic investigations.

1.2 OVERVIEW

Artificial images can be traced back to their generative models using deep learning techniques that rely on the data used to make those images. More advanced tools such as GANs and diffusion models allow users to make very realistic artificial images. This facilitates the creation and distribution of misinformation found online, identity theft, and the manipulation of digital media through edited images. Identities get stolen more easily. Concerns grow around misinformation, identity fraud, and editing in digital media. (Some experts worry about how fast these tools are spreading). The research focuses on two goals: first, whether features pulled from GAN encoders show distinct markers that could identify the original model used to generate a specific image; and second, whether deep learning systems can use these markers to quickly and accurately name the model behind a synthetic image. At least in theory, these features might offer a reliable path to attribution. It seems possible that some patterns in latent data are repeatable across different images. Arguably, the results could help detect when an image is artificially created. A key point is that real-world use may vary from lab conditions.

To analyze images generated by a GAN, an encoder is used to develop latent feature representations of the image, representing a unique pattern in the latent vector that can be used to train classifiers to classify which GAN produced the image. A fresh look at this approach begins by pulling signature details from fake visuals. After that step comes the creation of several sorting systems, each tested on how accurately it tags an images origin Different metrics of reliability and consistency for performance of different types of GANs determine how well datasets generated from different GAN architectures can measure the reliability of the features latent to the source of each generated image. If signatures reliably point to specific models, attribution becomes feasible. A key condition is that the latent representations must consistently carry identifiable traits across different generations. The study assesses whether such methods can be used in image forensics to identify AI generated content. The use of this technology would provide improved means of controlling the sharing and usage of AI generated images.

1.3 CHALLENGES

1. Limitations of Latent Inversion: Implementing the system also raised a few issues, one of which was the limitation of generative adversarial network (GAN) inversion methods. While decoding an image into its latent expression by means of an encoder, the reconstruction of the original image is simply an approximation and mostly suffers from loss of details, which in turn casts the potential and power of the encoders in a different light, as the very subtle characteristics of the GAN generator that are necessary for the GAN generator's image attribution may be hidden behind the encoders, especially since an encoder trained on a given GAN architecture might not work well with images generated from different GAN architectures, thereby further increasing the variability of the latent representations obtained.

2. Difficulty in Separating Model Fingerprints from Image Content: Another design problem stemmed from the necessity for the attribution model to concentrate on the generative fingerprints rather than the semantic content of the image. Most of the deep learning models are capable of mistakenly learning correlations with the subject matter, texture or visual aspects of the image, rather than with the intrinsic artifacts of the generative model that created the image at the initial rendering stage. Thus, to design the system that can extract the fingerprints independent of the content, an iterative design method involving architectural design and feature extraction strategies was used. This way, the classification model could learn the structural characteristics linked to the generative model instead of through the semantics of the image.

3. Computational and Resource Constraints: Image pre-processing, latent inversion and feature extraction operations on a large number of GAN-generated images consumed a very significant amount of computational resources up to the implementation stage. Running these operations repeatedly during experimentation had to impose a very large limitation in hardware resources usually in the form of GPU Memory and Processing Power. Besides during the implementation stage a lot of different problems popped up with environment configuration e.g. compatibility of different deep learning framework

versions, CUDA version and associated library dependencies; hence, a lot of debugging and configuration was done.

4. Cross-Architecture Latent Representation Alignment: Generative Adversarial Networks (GANs) have been evolving quite rapidly and every new version, for example ProGAN StyleGAN2, and StyleGAN3, comes with a different set of features. This made it challenging to analyze images from multiple architectures within a unified attribution framework. Mapping images from these models into a comparable latent representation space required validating that the extracted features still preserved architecture-specific characteristics. Designing features that could consistently capture these differences across architectures was therefore a significant challenge during system development.

1.4 PROJECT STATEMENT

This research project looks to develop a deep learning framework to trace the origins of synthetic images back to their original generating model through a feature representation in the model's latent space. Rapidly advancing generative models such as Generative Adversarial Networks (GANs) and diffusion algorithms are producing synthetic images that are gaining in popularity. Concerns have arisen over digital forgery, AI misuse and misinformation with GANs. The main objective of this research will investigate the possibility of utilizing features located in the GAN encoder's latent space as identifiers for the type of generative architecture used to create a specific image. Different GAN architectures will be used to generate images, and the images will be classified using latent features extracted from the image and machine learning techniques to identify the architecture that produced these images. Results will comprise classification performance comparisons for images generated from different models and image datasets. Performance metrics distribution across multiple model and image datasets will be the basis for evaluating latent space-based attribution methods' effectiveness and determining if these tools can be scalable in digital media forensics and increasing the accountability of GAN systems.

1.5 OBJECTIVES

1. Design a deepfake attribution model based on deep learning, which leverages the analysis of latent feature representations of GAN-produced images for attributing the generative model that created the synthetic images.

2. Study the latent feature representations derived from the GAN encoders to determine if these representations possess variations in their generator-specific features that may enable precise attribution of synthetic images.

3. Generate organized and annotated benchmark datasets of multiple GAN-generated image architectures (ProGAN StyleGAN2 StyleGAN3), so that attribution models can be developed and tested on a shared data source.

4. Implement a dedicated model training to test whether latent signatures can be used to accurately classify the originating GAN architecture through ML classification models with latent feature representations as the primary input.

5. Generate attribution result data for multiple generative models in order to assess the strength and reliability of classification results across different generative architectures; compare these with the classification accuracies of test data and the capability of generalization of different GAN-derived images.

6. Demonstrate that digital forensic science's latent space-based attribution methods could be innovative tools resulting in enhanced traceability, accountability, and detection of synthetic media generated by modern AI generative systems, including forensic watermarking in latent space and semiotic feature-based forensic scientific analysis as latent space-based forensic techniques.

1.6 SCOPE OF THE PROJECT

This research aims to find out if a deep learning-based approach (using the latent-feature representations of generated images) can identify which generative model made a particular synthetic image. The fundamental question of this work is if the latent signature(s) (latent features) obtained from the GAN encoder(s) will possess distinctive characteristics that can enable the accurate attribution of the underlying generative architecture that produced a given synthetic image. In order to solve this question, we will first create a dataset consisting of synthetic images generated by different GAN models (ProGAN, StyleGAN2, and StyleGAN3). We will then encode the synthetic images and use these encoded images to train a machine learning classification model to recognize the generator-specific patterns that exist in the latent space. Finally, the attribution framework will be tested (based on classification accuracy and robustness) to see whether latent-space representations can serve as effective tools for uncovering model-specific signatures that can be used for the attribution of synthetic images.

This research mainly concentrates on GAN generated images and latent space analysis for exploring Generative AI (instead of covering all types of generative AI, e.g. text-to-image and video). It was carried out compilation with classical computers in a deep-learning environment but, it focused only on using encoder-based latent feature extraction/classification techniques (not on making direct changes to the generative models). The paper addresses the matter of attribution evaluation primarily using well-known GAN data; therefore, the question of how the system can generalise to completely different generative models remain an area for further research. In spite of these limitations, the paper is an attempt to present latent representation analysis as a scalable and practical methodology for digital media forensics.

Chapter 2

Background

2.1 LITERATURE SURVEY

According to Chopade et al. (2025) "A Comparative Study of Pattern Based Methods and Hybrid Architecture For Image Source Attribution Of Deepfake Images", many researches have mainly focused on detecting whether the media is fake or not however very few have been dedicated to figuring out what model or tool was used to create the manipulated images. As a result, authors brought up a Hybrid Architecture known as DeepNoiseNet which identifies the sources of deepfakes by combining forensic noise patterns and semantic deep learning features. The classification approach proposed in this paper is aimed at strengthening and increasing the dependability of digital media forensics activities such as cybercrime investigation, misinformation detection, and media authentication.

Chopade et al. (2025) also reviewed earlier techniques used in deepfake analysis. Frequency-based methods (analyzing the frequency spectrum and examining anomalies caused by GAN-generated images) typically relied on the use of transformations such as Discrete Cosine Transform (DCT) and Discrete Fourier Transform (DFT) for their deepfake analysis methodologies. Additionally, researchers attempted to determine the authenticity of deepfake-generated images using analysis of residual noise and compressibles (such as Photo-Response Non-Uniformity [PRNU] patterns and Error Level Analysis [ELA]) caused by a manipulation operation. Researchers have evaluated custom made structural features (e.g., Local Binary Pattern [LBP], Histogram of Oriented Gradient [HOG], Gray-Level Co-occurrence Matrix [GLCM]) to uncover texture and structural discrepancies between the original and manipulated images generated by GAN's.

One of the major developments of deep learning revolved around the use of highly specialized Convolutional Neural Network, Vision Transformers and other future methods that help automatically learn features of discrimination with which deepfakes can be detected and

their source traced back (Chopade et al. 2025). The authors note that transfer learning and pre-trained models (such as EfficientNet) have enhanced features of robustness to image distortion and compression artifacts. Besides, several works have been combining features of various scales and resorting to ensemble learning techniques to achieve better performance over different generative sources; yet a lot of the systems still rely on single modality features and fail to generalize well across various datasets and/or lack robustness against post-processing operations (such as JPEG compression).

To address these limitations, Chopade et al. 2025 have come up with a new method named DeepNoiseNet that is a dual-cut architecture with two input streams. In one input stream, semantic features are extracted from EfficientNet and Vision Transformers, and the other input stream is formed by forensic noise feature extraction methods through PRNU residual analysis. These two input streams are then combined via an Adaptive Attention Fusion method that allows the model to utilize both high-level visual semantics and low-level forensic traces simultaneously for deep tracking tasks. Testing the model's performance not only showed a major improvement in the accuracy of deepfake source classification against previously tested techniques but also showed a noteworthy capability in handling compression artifacts and noise during the test phase.

- Advantage: The hybrid DeepNoiseNet architecture combines deep learning semantic features of images containing human faces (or faces in images) and forensic noise characteristics associated with these images, which makes it possible to accurately perform deepfake source attribution, besides being robust to compression artifacts and post-processing operations.

- Disadvantage: The deepNoiseNet dual-stream architecture may be resource-heavy and it might degrade in performance for detecting deepfakes generated by untrained models.

- Methodology: DeepNoiseNet's dual-stream architecture combines EfficientNet and Vision Transformers for semantic feature extraction from images while employing PRNU-based residual analysis for forensic noise feature extraction. The two types of

features are then combined via an Adaptive Attention Fusion technique and classified by a transformer-based classification model (Chopade et al.2025).

- Conclusion: Results from the experiment prove that the mixture of semantic and forensic features yields higher accuracy and robustness in the classification of deepfake sources. Besides, this outcome confirms the practicability of digital media forensics solutions as an effective tool.

Recently Zhang et al. (2022) come up with a deep learning method to identify which generation model has created a synthetic image in their work "Deep Learning-Based Synthesized Image Attribution Using Frequency Distribution Information". Authors highlight that the fast development of deepfake technology by Generative Adversarial Networks (GANs) makes it necessary not only to detect fake images but also to give specific attribution of them to their respective generation model(s). It is common for the different generative models to produce unique artifacts (i.e. fingerprints) that the authors claim can be used to trace the origin of the generated image.

Zhang et al. (2022) remark that almost all the previous studies on the identification of synthetic images rely on pixel level pattern analysis. Meanwhile, the researchers find out that different generative models imprint distinct frequency level patterns on every image they make. This is done by applying a transformation, e.g. Discrete Cosine Transform (DCT), which yields a periodic frequency spectrum; hence, all the generated images share the same recognizable signature that reveals the generation model. Contrarily, a good number of the previous frequency-based methods result in low accuracy attribution when photographs of different content to the training ones are compared.

To address these limitations, new deep learning framework was presented by Zhang et al. (2022) that introduces a learnable coefficient layer to the frequency spectrum of the input images. The proposed method will, on one hand, mask the frequency components that contain the content and, on the other hand, highlight the frequency components that correspond to the artifacts of the generative model. Aided by this enhanced frequency domain representation, a

classification module carrying out the ultimate task of determining the synthesizing model is given the input. Moreover, the design incorporates a Convolutional Block Attention Module (CBAM) for the purpose of helping with the selection of key frequency features at the time of training.

In an experiment using the Attribution88 dataset, Zhang et al. (2022) were able to evaluate their methods on a set of real images and synthetic images obtained from several generative models of common object types. The authors' new approach has been pitted against the present attribution methods which include RepMix and also the previous frequency based ones. Generally, the outcomes indicated that the new method fares similarly to all of these known object types but has attained better accuracy on all of the novel object types that were not in the training data, thereby demonstrating that this new method can generalize better to assigning synthesized images than the existing methods.

- Advantage: By deploying a learnable coefficient layer to operate in the frequency domain, the proposed method is able to efficiently expose the fingerprints of generative models, leading to a significant improvement in attribution accuracy and also a better capability to generalize to new and different image contents.

- Disadvantage: This method is bound to datasets with only low-resolution images and that will not have the ability to completely understand the generative model fingerprints for different image sizes and generation techniques unless the datasets are bigger and more diverse.

- Methodology: The system transforms input images to the frequency domain by applying the Discrete Cosine Transform (DCT). Secondly, it utilizes a learnable coefficient layer with attention mechanisms to highlight generative model artifacts.Finally, the modified frequency spectrum is passed to a deeo learning attribution module which performs the classification (Zhang et al. 2022).

- Conclusion: The research proves that examining the frequency distribution information along with deep learning can be a very effective way to trace back the synthetic image to the generative model that created it, and it also surpasses the existing frequency-based attribution methods especially when it comes to dealing with images that have not been seen before.

In a recent study, Pontorno et al. developed DeepFeatureX-SN as a deep learning framework for enhancing the accuracy and generalization ability of deepfake detection systems. With deepfake media characteristics becoming increasingly realistic due to recent advances in generative AI technologies like GANs and diffusion models, detecting media of this type has become increasingly difficult to achieve. With the production of highly realistic synthetic media (deepfakes) becoming possible due to recent breakthroughs in generative AI technologies such as GANs and DMs, it is getting progressively challenging to detect such media types. Pontorno et al. point out that the rise of these novel generative architectures renders current detection methods non-generalizable because the images differ to such an extent from those on which the detectors were trained. This is why the authors came up with a completely new model, which apart from recognizing real versus fake, can also identify if the image was made using GANs or diffusion models.

Researchers have mainly investigated the detection of deepfakes via the examination of artifacts that are inherent to the generated images from the original deepfake generator. For instance, they have detected color channel inconsistencies, have performed Discrete Cosine Transform (DCT) in order to reveal abnormal frequency patterns in an image, and have identified artifacts created due to the up-sampling processes in generative networks. Many deep learning methods have also been suggested by using various architectures (ResNet, CNN-based type detectors). Even though the current deep learning tools produce quite accurate results when tested in standard setups, they are usually less effective when tested on deepfake images made from new architectures and also subjected to real-world manipulations like compression and geometry alterations.

The aforementioned knowledge gaps have motivated Pontorno et al. (2025) to propose a fresh model named DeepFeatureX-SN through a tripartite architecture comprising three specific base models aimed at identifying let's say unique characteristics within the real images, GAN images as well as diffusion model generated images. All of the three base models are trained via Siamese neural networks with contrastive learning to gain feature mappings in which like-images are not only mapped nearer to each other but also unlike-images are mapped farther apart than others within the triangle of similarity space. Next, the feature set from each of the three base models is concatenated together into one feature vector and then passed through a CNN-based classifier for final image classification.

Pontorno et al. conducted a series of second-order experiments with multiple types of image-generating techniques at the same time employing a very large dataset. There were many GANs and diffusion architectures along with real images of these generative techniques. The newly developed model did 97.29% correct predictions on a three-class classification problem and realized superior generalization performance not only on generative techniques that were part of training data but also on those that are completely new to it. This research has also led to advancement of contrastive learning technique along with the ability of the feature extraction module to identify minute details resulting in an enhancement of the robustness and flexibility of deepfake detection systems by a significant margin.

- Advantage: DeepFeatureX-SN is an innovative architecture for detecting Deepfake images, whereas Contrastive Learning helps to extract multi-feature and allows zero-shot detection of Deepfake images that are produced in any way of a person, e. g. Images made using Unseen GAN models.

- Disadvantage: Large-scale deployment of the proposed system in environments with low resources may be limited by the dependence on multiple feature extracting models that are computationally intensive and the development of complex training protocols.

- Methodology: For Real, GAN, and Diffusion image types, SNN's will first be able to identify the features that are the most distinguishable and create three different base

models that will be used later for feature extraction.The three individual feature embeddings will be combined and used as inputs to the CNN in the image recognition process (Pontorno et al. 2025).

- Conclusion: From a theoretical perspective, DeepFeatureX-SN is more accurate than other existing deepfake detection tools and aligns better with reality. The growing capability of AI architectures resulting in an abundance of different types of synthetic media, among other factors, will most probably ensure the continued relevance of DeepFeatureX-SN deepfake detection for a long time.

Uhlenbrock et al (2026) conduct research regarding the structural characteristics of contrastive language-image pretraining feature spaces. While the focus of their article, "Latent Landscapes: The Topological Structure of CLIP Feature Space for Detecting Synthetic Images", is on the relationship between language and image based generative models (e.g., GANs, Diffusion Networks), the authors note an issue with increasing levels of photorealism through these types of models. It is now possible to generate an image using these types of methods (or manipulate an image) and to create a digital image that looks realistic; therefore, an important consideration going forward will be to develop approaches for detecting whether a digital image was generated or manipulated using current methods. Using their research, the authors suggest that most detection techniques designed to detect synthetic images (i.e., network-based detection techniques) are likely to be ineffective in detecting images produced by newer generative networks or images that have been altered by post-processing (e.g., compression or resizing).

In their study, Uhlenbrock and co-authors (2025) investigated the organization of both real (natural) images and synthetic (artificial) images in the CLIP latent feature space to find out if these latent representations depict enough information on how to tell natural images from artificial images. The authors reduced the dimensions of high-dimensional CLIP embeddings to the wells of low-dimensional visualizations, observations from which show that natural and artificial images occupy separate regions in the latent space. Nevertheless, the authors discovered that the training set of the detector substantially influences the shaping of these

distributions since classifiers trained on one set of samples may not perform well in the dark on messages produced by a sampling method. Hence, geometrical properties of the latent space should be considered as a matter of priority when one is developing robust detectors.

To evaluate the concept of robustness in more depth, Uhlenbrock et al. (2025) studied the effects of several post-processing operations such as resizing, the addition of Gaussian noise, JPEG compression, histogram equalization, contrast enhancement on the movement of image points in CLIP's feature space. The results of the study indicated that post-processing operations could change an image's feature representation by different distances and sometimes even push them closer to or beyond a decision boundary between real and synthetic images. Following the study's findings, the authors decided that because synthetic and real images react differently to a set of transformations, this will give rise to some regular patterns which can then be used to enhance discrimination abilities.

Based on the observations of Uhlenbrock et al. (2025) revealed a succinct manner of encoding features respective to image shifts occurred due to post-processing operations. Their approach was based on CLIP features coupled with a couple of extra metrics meant to determine distances and directions of the shifts induced by post-processing transformations. These features became compatible with Support Vector Machine (SVM) which is known to enhance the robustness and generalization of the model's performance when trained on various datasets. The findings showed that the overall detection accuracy significantly went up by using CLIP features and post-processing features together for training, at the same time, providing the best resistance to the new generative models.

- Advantage: It is a great benefit that the researchers obtained a profound understanding of the geometry/topology of the CLIP feature space. This understanding will pave the way to tie synthetic image detection with higher reliability and flexibility.

- Disadvantage: Some generative models or datasets might be restricted from use as their feature distributions could differ from those of the model's training dataset.

- Methodology: The team first compared the CLIP representations of real and synthetic images. Then, they examined how post-processing affected these features.Based on the changes in distance and direction, a feature vector was formed. The feature vector was then used to train an SVM model created by Uhlenbrock et al. 2025 that was capable of classifying the features.

- Conclusion: Recognizing modifications from post-processing and having a grasp on topology-based spatial mapping can lead to increased reliability and flexibility of synthetic image detection systems.

The attribution framework MAID (Model Attribution via Inverse Diffusion) is developed by Zhu et al. (2025), is an approach to identify which model has generated a certain image via attribution precisely through the use of inverse diffusion equations. Generative models such as GANs (Generative Adversarial Networks) and DMs (Diffusion Models) have flourished and made it extremely easy to produce highly realistic 'deepfake'-style images, leading to numerous unfounded fears including the spreading of misinformation, fraud, and the alteration of images. Furthermore, the authors mention that the majority of attribution techniques perform poorly on images created by multiple generative models or rely on extra data, for example, the sampling prompt and model parameters which typically may not be available."

Zhu et al. (2025) identify two principal problems associated with existing attribution methods. First, traditional attribution methods generally depend on GAN images as the only source of generated images. Besides not being extendable to diffusion-based generative models, they also fail to consider the vast array of diffusion-based generative models that have become available due to technological advancements. Second, several recent attribution methods heavily rely on white-box level information, such as model weights, text prompts, and semantic metadata to facilitate the attribution process. Since none of this information is available during normal forensic investigations, the capabilities of these methods as presented in previous publications have been quite limited. Hence, the writers of this article have proposed an attribution method that can operate independently and independently from the image creation framework and can reliably find the source model of a generated image just by looking at the

generated image without any other information.

To assist with this task, Zhu and colleagues proposed MAID (2025) . MAID utilizes distributions already trained to extract image features via an Inverse Diffusion Process, treated as a Denoising Autoencoder, in which noise characteristics of the two types of characteristics in the Diffusion Model are studied, termed Diffusion Model Activations (DMAs). DMAs have the ability to identify and deliver information about the unique patterns that are present in images made by different types of models. Next, the DMAs are considered as input features for the classifiers to predict the source model of an image, thus allowing attribution of both GAN-generated images and those that are made with diffusion.

Zhu et al. (2025) carried out experimental evaluations using DiffusionForensics and Artifact which consisted of images generated by a variety of different generative models. The results showed that the MAID framework greatly outperformed conventional attribution frameworks not only in accuracy but also in robustness. For example, on the DiffusionForensics dataset the MAID approach achieved an attribution accuracy of approximately 97.76% in figuring out the original source of AI-generated images. Besides being highly accurate in pinpointing the original source of AI-generated images, the MAID framework also exhibited significant robustness to image distortions such as Gaussian noise, JPEG compression, and blur.

- Advantage: The MAID framework can be considered a nearly universal model attribution approach with diffusion models being used as feature extractors to identify images generated by both GAN and diffusion models without any additional information besides the image itself.

- Disadvantage: The idea underlying this method is to use pre-trained diffusion models for inverse diffusion process. Due to the high computational cost and complicatedness of the implementation, this may be considered a drawback.

- Methodology: A trained diffusion model is used to perform inverse diffusion on an image to retrieve Diffusion Model Activation (DMA); these are the model-specific

signature features which in turn are employed for training a classifier (eg. ResNet-50) to identify the source generative model (Zhu et al. 2025).

- Conclusion: Findings obtained from the experiments demonstrate that MAID outperforms other existing methods in terms of the accuracy and robustness of attributing AI-generated images to their source generative model, and thus, it can be regarded as a suitable framework for conducting forensic investigations on synthetic media produced through this methodology.

Guarnera et al. (2022) conducted a study to determine if tracing back the specific generative model that created a deepfake image is feasible, other than merely detecting the deepfake (fake) itself. They pointed out that most of the works have been focusing on deepfake detection or the classification of different GAN architectures. Only negligible academic work has been considering how the creation of an image is tied to a particular model, as per the researchers. Nonetheless, pinpointing these instances will be essential to combatting intellectual property theft and ascertaining the source of synthetic media in digital forensics. Besides, they reported that every GAN produces a distinct latent space and parameter set. These two features become the foundation for model attribution.

Guarnera et al. (2022) explored the fundamental properties of the latent space of StyleGAN2 models. They also wanted to see if small changes in the parameters of the two models could lead to each model generating visually different images. To carry out their study, the authors first made a dataset of images by slightly changing the training parameters of multiple StyleGAN2 models. Their experiment pipeline indicates the process starts with the creation of an image dataset, then an encoder is trained to identify images generated by different models, and lastly, metric learning methods are employed to analyze the representations of generated images and thus determine the model attribution.

Originally, this technique was introduced by Guarnera et al. (2022) who used a ResNet-18 encoder to identify images produced by 50 different StyleGAN2 models. The intended function of the encoder is to derive, from the images, distinctive discriminative features so, that these

features are used to come up with a similarity metric that is then used to determine whether two images belong to the same model. Besides, they proved that the encoder was able to discriminate the images generated from the 50 known models with an overall accuracy of about 96.24%, which proves that although to the naked eye the images may look almost identical, in reality, each one is characterized by distinct patterns formed as a result of the generating process.

In the research work by Guarnera et al. (2022), some individual tests were done as a continuation to check if their method really is robust so they also included some perturbations such as Gaussian blur rotation mirroring, scaling, and JPEG compression. The findings revealed that the performance, although changed, was still very good overall metrics of performance of about 81% accuracy while under conditions of noise or distortion. The authors argued that model-specific traces in generated images can be exploited for forensic attribution purposes, and their proposed pipeline is a great tool for forensic investigators to identify the source of synthetic media.

- Advantage: The new method reveals that each image source leaves behind a unique 'fingerprint' (ie. recognizable patterns) that can be used to link deepfake images to their original sources even when they undergo significant changes visually.

- Disadvantage: For effective classification of unfamiliar or heavily altered images (ie. "new architectures") the system has to be initially fed with images of well-known models.

- Methodology: The paper's authors created images with a number of fine-tuned StyleGAN2 models, used the ResNet-18 encoder for a correct source model classification of the images they were given, obtained the latent features, employed metric learning on these latent features to compare them with one another and thus see if they were generated by the same source model (Guarnera et al. 2022).

- Conclusion: The experiment results suggest that even the most subtle traces hidden

within GAN images are enough to pinpoint the model that created them, and as such this work will serve as an excellent basis not only for further studies of model prediction but also for model forensics of digital assets.

Dell'Anna et al. (2025) reported that TrueFake is a collection of various deepfake video segments that serve the purpose of testing the efficiency of deepfake detection systems in authentic scenarios. Over the last few years, the advent of Generative Adversarial Networks (GANs) and Diffusion Models (DMs) has facilitated the public's access to generative media. Besides, not only images generated with GANs and DMs but also other kinds of generative media are becoming increasingly accessible. Moreover, the proliferation of such kinds of images has resulted in a growth of social media use and it is expected that this trend will continue; however, sharing images in this way has its dark side too because it may lead to identity theft, misinformation, and online fraud. Besides, the authors pointed out that since humans are often fooled when looking at real and AI-generated images, the role of automated deepfake detectors is bound to increase gradually.

DellAnna et al. (2025) point out that most deepfake detection methods that exist today were primarily created and tested in a lab environment, with many methods relying on a clean dataset. In fact, in the real-world scenario, images incidentally are mostly shared through social media platforms such as Facebook, X (formerly Twitter) Telegram etc. A lot of adjustments happen when users share an image via social medias such as compression, resizing, and other processing functions. These changes can even lessen the signs that detection models are reliant on to identify the image. Readers may find that according to the authors processing at the level of the platforms can cause a great loss in the capability of fake image detectors because it hides the subtle signs that can be found in synthetic images.

The TrueFake Dataset was created by Dell'Anna et al. (2025) to fill this gap in the literature. It is composed of around 600k images, with the same number of real-life and artificial samples. The dataset includes roughly 100k images from the FFHQ and FORLAB datasets, 200k images generated by diffusion algorithms like Stable Diffusion and FLUX, and 120k images obtained using GAN architectures such as StyleGAN. The rest of the 180k images have been distributed

through social media platforms (Facebook, X [formerly known as Twitter], and Telegram) by APIs that simulate the way real media are distributed, giving researchers the opportunity to investigate how social media compression and processing affect detection performance.

In 2025, DellAnna and colleagues also reviewed several state-of-the-art techniques for detecting media (including CNN as well as CLIP models). They found through their experimental studies with social media-processed (or "accidentally altered") image data that a large part of the image detection approaches, especially those relying primarily on local pixel artefacts for image item identification, experience a drastic drop in performance after image processing by social media platforms. This evidence points to the importance of engendering detection systems that are not affected or are robust against transformations enacted in the real world, such as compression, resizing, and the distinctive image processing methods carried out by various classes of social media.

- Advantage: The TrueFake dataset was aimed at creating a very big and accurate benchmark that can be used to test deepfake detection tools with real-world scenarios. The dataset imitates images processed through social media platforms like Instagram and Facebook.

- Disadvantage: While it is very thorough, the TrueFake dataset is only a snapshot because the landscape of generative models is rapidly changing and also social media platforms are using sophisticated ways to process images. Therefore, the dataset will have to be updated regularly.

- Methodology: In order to mimic real-world conditions, the authors of the paper have put together a dataset consisting of real, GAN and diffusion generated images and have shared a portion of these images through social media platforms.Different detection models are also going to be tested on both non-socially processed and socially processed images (DellAnna et al. 2025).

- Conclusion: The paper indicates that image processing through social media drastically

changes the performance of deepfake detection models that are widely used at present. Therefore, it underlines the necessity of testing forensic methods with datasets that truly represent the environment in which images are shared.

Fan et al. (2024) claim that the way images are compressed during a transfer over a network affects the quality of attributing GAN generated fake faces. Generative Adversarial Networks (GANs) have produced so-called synthetic faces which are pretty much indistinguishable from real ones, and with further improvements, they can be exploited for unethical purposes such as identity abuse and misinformation, among others. Despite the fact that some fingerprinting methods have been introduced to identify the source of generated images, when an image is compressed using algorithms such as JPEG, the fingerprints can be destroyed, thus aggravating the attribution of images. To this end, the authors argue that upgrading the state of deteriorated GAN fingerprint might be very much necessary for better performing attribution of face fakes.

Fan and the other authors (2024) wrote that two categories of images are the ones that are affected by image compression: the first is that the image content can be visually degraded directly, and the second is that the GAN fingerprints embedded in the images can be degraded indirectly. Compression causes the addition of visual artifacts or the removal of very important details of images and, at the same time, it leads to a big decrease of the discriminability of GAN fingerprints. The majority of image restoration techniques available at the moment (CNN-based, transformer-based, diffusion-based, etc.) are mostly focused on restoring the visual quality of the image, working solely within the image domain to recover the lost image information, whereas GAN fingerprint information is located in the feature domain (which is separate from the image domain).

Fan et al. pinpointed the flaws and to resolve them, they came up with a new technique, the Dual-domain Multi-Model GAN Fingerprint Restoration framework (DMFR). DMFR is made up of two image restoration channels working simultaneously: one that repairs the images directly and another that takes care of the fingerprint ones. The channel of the image domain works at first on a typical image with compression then, it will help to better extract the GAN fingerprint(s). On the other hand, the fingerprint domain channel starts with extracting the GAN

fingerprints then, it will restore the fingerprints by applying the target restoration model. In the end, by using both independent but complementary domains, the suggested framework is aimed at yielding a better quality of reconstructed GAN fingerprints and ultimately, differentiating those fingerprints more effectively.

A novel multi-model restoration fusion method presented by Fan et al. (2024) uses a combination of various GAN-specific restoration networks to recover fingerprints with higher distributive accuracy compared with existing solutions. Different model results are combined with decision-level fusion networks to produce the final attribution results. Importantly, the tests demonstrated that the DMFR outperformed in terms of attribution accuracy and F1 scores against existing restoration-based methods. This result highlights the success of the dual-domain restoration concept at different levels of JPEG compression.

- Advantage: This technique achieves higher attribution accuracy by integrating fingerprint-domain restoration methods with image-domain restoration methods even to a level at which both use barely yield any results, due to image compression.

- Disadvantage: Employing multiple types of restoration models and fusion networks adds extra computational load to the system.

- Methodology: The dual-domain restoration architecture of the proposed system consists of two main restoration pipelines: one for images and one for fingerprints. Generating images during the dual-domain restoration processes of each image, all of those images eventually end up in the same multi-model restoration fusion network, which then reliably attributes the output image using the initial GAN model to produce the model output (Fan et al. 2024).

- Conclusion: According to the study, the application's ability of dual-domain restoration and multiple model fusion significantly contributes to the successful attribution of fake faces of restored GAN fingerprints compared to cases where the image is degraded due to compression during transmission.

Yu et al. (2019) investigated if one could possibly track any distinctive marks of a Generative Adversarial Network (GAN) through its production of synthetic image data. Thanks to the rapid progress of GAN's capability, extremely realistic synthetic representations have been created. Nevertheless, the issue of potentially harmful uses of this technology like in digital forensics, intellectual property protection and misinformation detection has raised concerns. Further, the authors noted that most "traditional" methods for detecting deepfakes rely on the visual features or discrepancies of the synthetic image. However, many of these methods do not guarantee reliable detection especially when models with different architectures from the traditional detection methods are used. Thus, the authors decided to carry out this research with the motive to find out if there were any special features or "fingerprints" of GAN synthetic images that could possibly be used to link a synthetic image created by a GAN to the same GAN model that generated it.

Yu et al. (2019) have stated that each GAN instance owns a unique model "fingerprint" which is determined by several factors including: (1) the architecture of the neural network, (2) the distribution of the data used to train the particular GAN, and (3) the initialization parameters for the particular GAN. These fingerprints are present in every generated image. When features from such generated images are transformed into a feature space, these features form clusters of distinct groups corresponding to the GAN model that generated that particular image. This fact implies that generated images contain latent structural features such that the images can be differentiated from their respective real counterparts as well as from the images produced by different GAN variants.

Yu et al. (2019) the authors of a work, created what is called an attribution network by building a deep convolutional neural network (CNN) classifier that aimed at figuring out the source of an image through its unique markings. The attribution network learns by considering source pairs (e.g. a celebrity's photograph), whereby each image is linked to the corresponding generating model. After the CNN has been exposed to this data set, it becomes capable of producing feature embeddings for all image fingerprints; in addition, it has also mastered the art of assigning weights to each model fingerprint. Then, by analyzing all of these associated

representations (i.e. classifying images based on the feature representations learned), the attribution network is able to decide if a particular image is either a genuine photograph or taken from some specific GAN model (Fake Image). The experiments conducted using CelebA, LSUN, and other images from different GANs (e.g. ProGAN SNGAN CramerGAN MMDGAN) datasets show how extremely the entire system is at distinguishing real images from fake ones.

Yu et al. (2019) demonstrated that the consistent fingerprints of GANs are present in a variety of factors such as image pieces in different frequency bands and/or image patches. They also tested how well those fingerprints can withstand perturbations (i.e. noise cropping, JPEG compression, re-lighting). Although these perturbations can weaken the fingerprint signal captured by the system, the researchers were able to restore performance by fine-tuning the model used to create the fingerprint. The experimental results showed that the method proposed far exceeded the accuracy of attribution provided by the traditional methods (e.g. k-nearest neighbours Eigenfaces PRNU-based fingerprint detection), thereby proving that the consistent fingerprints produced by GANS are indeed present.

- Advantage: The research shared shows that the fingerprints found in GAN-produced images are not only stable but also have unique characteristics, which makes it possible to identify the exact GAN model that was used to produce a particular image.

- Disadvantage: To create an attribution model, one has to provide training data with labels from known GAN models; therefore, the attribution system would not be able to identify, represent, or handle the Gamb/Images generated by completely new and unknown GAN architectures.

- Methodology: The paper employs a DCNN classifier that is trained on an image/source dataset to obtain and understand both image and model fingerprints so as to categorize an image into real or generated by any of the GAN models (Yu et al. 2019).

- Conclusion: This study suggests that all GAN-generated images bear recognizable and

unchanging fingerprints, and these fingerprints can be used for successful attribution of synthetic images to their generating models.

Hirofumi et al. (2022) investigated the problem of identifying the specific GAN model responsible for generating a synthetic image. With the increasing availability of powerful generative models, it has become possible to produce highly realistic images that are difficult to distinguish from real photographs. This capability introduces serious concerns related to misuse, including impersonation, copyright infringement, and misinformation. The authors emphasized that determining the exact GAN model used to generate an image is important for ownership verification and forensic investigations.

Dushi et al. (2022) conducted research on the method of determining the specific GAN Model that is responsible for the production of a synthetic image. The emergence of more sophisticated generative technologies has led to the creation of highly realistic images that can barely be distinguished from real photographs. This ability has triggered many concerns regarding the possibility of misuse such as impersonation, copyright infringement and spreading false information, among others. In their view, it is indispensable to identify the particular GAN model that was employed in the generation of an image not only for the purpose of verifying ownership but also for any forensic investigation that may follow.

Hirofumi et al. (2022) came up with a method of attributing images to their generating model by latent recovery as a solution to this problem. They make a fundamental postulate that if an image was created by a certain GAN model, then it should be possible to accurately reproduce that image with the generator of that very GAN model by optimization from latent space. However, since genuine GAN output is hardly ever created exactly as expected, their target is to find a latent vector that can be used (with the generator) to produce a reconstructed version of the input image. So, if the reconstruction error of the reconstructed image is less than a given size compared to the generated version, they assign the given generated image as having come from the particular generator. The authors conducted the assessment of the attribution method by examining the recovery success rate and verification thresholds for each model using recovery success rate data and reconstruction error distribution data, respectively.

Hirofumi et al. (2022) conducted the evaluation of their framework by employing datasets such as CelebA and LSUN, along with various GAN architectures like ProGAN and StyleGAN. The experiments indicated near-perfect verification capability (AUC close to 1.0) in numerous test settings and demonstrated that their method was very robust against perturbations including Gaussian noise blur JPEG compression, and model retraining attacks. These findings imply that latent recovery is capable of identifying if an image has been generated from a specific GAN model or not.

- Advantage: Our way of attributing a GAN that is already built and can work with pre-existing (publicly available) GANs will not need any kind of preprocessing of the generative GAN architecture.

- Disadvantage: Since latent recovery is optimization intensive, it would take extra time to analyze large sets of images.

- Methodology: The framework optimizes the latent variables of a potential GAN architecture to generate an image based on its model; thus, the model will give meaning to the reversed image. The reconstruction of the image will be done using the reconstruction error and also the average number of successful reconstructions of that type of image will be used to find the one which can generate the original image (Hirofumi et al. 2022).

- Conclusion: Latent recovery that is accurately and reliably traceable to the generating GAN models, as done in the cited study, implies that image generation attribution is certainly possible at a very high level of accuracy and reliability. Hence, this study offers both a conceptual and methodological framework for organizations, law enforcement agencies, etc. who want to enact post-hoc attribution of GANs from a forensic casework perspective.

Tariang et al. (2024) provided an in-depth at a variety of methods for identifying (detecting) and attributing (finding out who or which source originally made) synthetic images generated by latest generative models. Since generative AI technology like Generative Adversarial Networks (GANs) and diffusion models is advancing very fast and becoming better in producing synthetic images that look almost real, there is a mass production of such images. However, it should be noted that along with these improvements, which have already been applied in many ways offering benefits to different sectors, new threats related to misinformation, counterfeiting and manipulation have emerged. Hence, the development of dependable (automated) tools that can effectively separate (or differentiate) synthetic films/real ones is gaining momentum amongst researchers.

Tariang et al. (2024) argues that synthetic image verification is essentially a two-fold operation: detection - understanding if an image is real or artificially generated and attribution identifying which specific generative model created the synthetic image. Ultimately, for example, the generative models used to create synthetic images in the past could be detected by the visible artefacts of their generation (e.g. wrong shadow placement disfigurations etc.); but, contemporary generative models are so highly developed that these artefacts cannot be considered reliable signs of machine-generated images anymore (realistic enough so that no visible artefact is present). As a result researchers in most cases, rely on the forensic evidence that is not visible and the statistical patterns that have been left by the generation of the image to identify synthetic images.

Tariang et al. (2024) have summarized various techniques in their paper to identify the production of synthetic images. The methods can be basically divided into two main categories: data-driven (which rely on deep learning, including convolutional neural networks and transformer-based networks) and forensic (which trace low-level cues like frequency patterns, noise residuals, and model fingerprints in their databases). Image generation processes, which many times involve steps like upsampling and filtering, lead to the emergence of some artifacts in the frequency domain; these artifacts allow forensic analysis to figure out whether an image was produced by a computer or not.

Tariang et al. (2024) have summarized various techniques in their paper to identify the production of synthetic images. The methods can be basically divided into two main categories: data-driven (which rely on deep learning, including convolutional neural networks and transformer-based networks) and forensic (which trace low-level cues like frequency patterns, noise residuals, and model fingerprints in their databases). Image generation processes, which many times involve steps like upsampling and filtering, lead to the emergence of some artifacts in the frequency domain; these artifacts allow forensic analysis to figure out whether an image was produced by a computer or not.

- Advantage: The paper provides a comprehensive review of methods for detecting and attributing synthetic images, such as forensics methods, and deep learning approaches for detecting images produced by AI.

- Disadvantage: Most of the present detection and attribution methods fail to generalize across generative models that are new and therefore unknown and also tend to show poor performance for images that have been compressed, resized or photoshopped in any possible way in the real world

- Methodology: Authors surveyed literature and experimented to identify which among different techniques can detect and attribute images generated by AI. Some of the methods they used are convolutional neural network (CNN) based classifiers, transformer models, frequency-based forensic analysis, extracting artificial fingerprints, and attributing images through inversion using generative adversarial networks (GAN) and diffusion models (Tariang, et al.2024).

- Conclusion: The authors state that detecting and attributing synthetic images with effectiveness is possible in a closed environment using current technology however new methods must be developed so that they can be applied to a broader range of generative models and can recognize and attribute to an open set of synthetic images created by AI in the real world.

Yu etal (2021) showed the use of artificial fingerprints embedded in the generative model to create proactive deep fake detection and source attribution. With the fast progress of Generative Adversarial Networks (GANs), the making of synthetic images that look like photographs is getting more and more credible since it becomes very difficult to detect them by humans alone. The traditional types of deep fake detection usually depend on the detection of the artefacts that are left over (or the inconsistencies) after the generative model has been created but as these methods are being enhanced, the artefacts will also continue to improve and become less and less detectable over time which would make the majority of these traditional ways of detecting deep fake content quite ineffective (Yu et al 2021).

Yu et al. (2021) suggested to the solution of this problem the idea of placing fingerprints inside the training data of generative models. These fingerprints are transferred to the final trained generative model as it produces the images in the same way and hence all images generated from the generative model carry fingerprints. Essentially, this approach relies on steganographic methods that enable a user to put some hidden information inside the original training images without greatly affecting the quality of the image. After the generative model is trained with these fingerprinted images, the model understands the hidden fingerprints implicitly and therefore is able to include the fingerprints in each synthetic output image and correctly identify the synthetic image with the particular generative model that was used to create that synthetic image.

The framework proposed by Yu et al. (2021) consists of multiple stages, including fingerprint embedding in training data, generative model training using the modified dataset, fingerprint decoding from generated images, and fingerprint matching for attribution. Experimental evaluation demonstrated that these fingerprints are transferable across various generative models such as ProGAN, StyleGAN, StyleGAN2, and BigGAN. The results showed high fingerprint detection accuracy while maintaining the visual quality of generated images, indicating that the embedded fingerprints do not significantly degrade generation performance.

Yu et. al. (2021) also took a look at the resilience of artificial fingerprints in the face of different disruptions/noise compression cropping blurring etc. The experiments showed that

even after multiple post-processing operations on the images, fingerprinting could still be detected. Moreover, the method hit accuracy of nearly 100 percent in detecting deepfakes and in assigning them to the right source across several datasets. This comes across as one of the strongest evidences that artificial fingerprints may be employed as a traceability tool in the creation of synthetic images.

- Advantage: It is possible to incorporate artificial fingerprints into generative models' training which enables the identification of which generative model produced an image; hence, this technique is not only proactive but can also contribute to the establishment of provenance for deepfake detection and attribution.

- Disadvantage: Because generative models have to be trained with the embedding of fingerprints, this technique will be ineffective for generative models that were not fingerprint-trained.

- Methodology: Deep learning-based steganography is used to embed fingerprints into the images. The embedded fingerprints are then passed from the image to the generative model during training and finally decoded from images generated by the generative model for deepfake detection and the attribution of deepfake images to specific generative models (Yu et al. 2021).

- Conclusion: Studies reveal that artificial fingerprinting is a powerful method to carry out deepfake detection and attribution, capable of raising the detection accuracy and production quality to a high level while still supplying a method of tracing the origins of synthetic media.

Sun et al. (2022) explain that FakeTracer is a pre-emptive defense mechanism against highly realistic deepfake photos . It does so by changing the way the deepfake model learns to create images. Generative models like Generative Adversarial Networks (GANs) and Variational Autoencoders (VAEs) are quickly evolving and becoming very capable of producing realistic avatars. Deepfakes first generate a face image of someone else, then they

use this image to extract the face attributes, which they finally use to re-construct the face using a decoder network. The faces produced by deepfakes can be so realistic-looking that they can be used in very harmful ways, which can include creating false information, impersonating someone else, and providing digitally altered images.

Sun et al. (2022) state that deepfake detection countermeasures can be broadly classified into two big categories: passive defense (without control over deepfake development) and proactive defense (with control over deepfake development). Passive defence methods mainly focus on recognizing the faults and/or inconsistencies inside a synthetic image with the help of machine learning algorithms, which are trained to differentiate between a genuine and a counterfeit image. The negative aspect of passive defence methods (for instance, they offer the possibility of securing one level higher against deepfakes) is that they do not give any measure about how deepfake models can be trained/developed (in other words, it is technically impossible to put any kind of limitation on the development of deepfake models); besides that, in most cases, they become less effective when generative technologies evolve. On the other hand, proactive defence methods take a stand while deepfake models are being trained or generated in order to produce synthetic images that are easier to be detected.

To implement this idea, Sun et al. (2022) introduced the FakeTracer framework, which embeds two types of traces in training images: Sustainable Traces (STrace) and Erasable Traces (ETrace). These traces are added to facial images before they are uploaded to social networks. When malicious users collect these images and train deepfake models using them, the training process becomes contaminated. As a result, the generated deepfake images retain the sustainable traces while the erasable traces disappear during generation. This characteristic enables investigators to identify synthesized images by detecting the presence or absence of these traces.

Sun et al. (2022) conducted an assessment of the Celeb-DF dataset, among others, to decide the excellence of the suggested technique for the recognition of deepfake images and deepfake videos. The laboratory outcome reflected that FakeTracer is a great tool to uncover the hidden use of deepfake images, though it also keeps the average quality of the image at a reasonable

level. Using the recommended method, the precision in detecting bit-matched traces was very high, and the soft spots to changes (e.g. JPEG compression; Gaussian noise; blurriness) were very few. It can be inferred from these data that active manipulation of the training data may be one of the ways leading to more trustworthy detection of deepfakes compared to the levels achieved with passive standard detection methods.

- Advantage: The FakeTracer system introduces a brand new method of defense against deepfakes by poisoning the training data. Since the whole process depends on the image data, the produced model is eventually able to generate fake images that contain identifying traces. This means that it is possible to detect synthetic images with very high accuracy. An excellent advantage is that a single contaminated image can cause both a huge positive impact on detection and a very tiny negative impact on the generation quality of that image.

- Disadvantage: One of the main drawbacks of the FakeTracer system is that it can only identify and classify the deepfakes that are specially processed with a trace before being uploaded and/or trained with image data. This, in turn, makes it much less effective against the existing or even pre-trained data sets.

- Methodology: The authors of this article (Sun et al, 2022) make use of a trace generation network to embed in them both sustainable and erasable traces in facial images. These traces act to change the training process of deepfake models so that any synthetic images produced not only contain sustainable traces but also do not contain eraseable traces. This leads to the capability of detecting synthetic images through trace interpretation.

- Conclusion: Planting poisoned samples within training datasets turns out to be a highly efficient way of proactively defending against deepfake images. The quantity of contamination does not lead to a significant fall in the quality level of the base images, whereas detection accuracy is at a very high level.

Kim et al. (2024) WOUAF (Weight Modulation For User Attribution And Fingerprinting) is a method for identifying who is responsible for an image generated with text-to-image diffusion models (e.g. Stable Diffusion) which was introduced by Kim et al. (2024). It is becoming increasingly accurate to generate photo-realistic images from text-based descriptions by using generative models. Generative models have seen explosive growth over the past several years. Although this creates many new opportunities for creativity, there are also many valid concerns about the challenges the advancement of generative technologies will create, including the proliferation of fake news, manipulated images, and other emergent issues. Consequently, the creation of an accountability and traceability system for all generated content will be essential moving forward.

According to Kim et al. (2024), conventional fingerprinting techniques focus on the use of peripherals/modules that embed identifiable information onto the images as the main fingerprinting method. However, these peripherals/modules are quite vulnerable to user manipulation or removal, especially in open-source environments where many users have these methods. Thus, the authors suggest another solution WOUAF which is a distributed-centered approach, and that will insert the individual's unique fingerprint into the internal model's parameter set so that each image will have the user's unique fingerprint, thereby enabling the user to identify not only which user generated that image but also whether that image was created by that user.

By weight modulation at the input, fingerprints may be fed into the Stable Diffusion model (Kim et al. 2024) by changing the fingerprint representation in such a way that the weights which are used by the diffusion model decoder can be modulated. The set of weights is then tuned, and another neural network, called the decoding network, is trained to recognize the embedded fingerprint from the images produced by the diffusion model. As a result, there are three stages in the entire process of encoding fingerprints not only, but also creating fingerprint images via weight modulation and decoding the fingerprint images for identification.

Kim et al. (2024) experimentally verified that the proposed WOUAF method led to nearly perfect attribution accuracy and, at the same time, enabled the creation of good quality images.

Based on experiments with datasets such as MS-COCO and LAION-Aesthetics, the researchers observed that the FID and CLIP-score metrics of WOUAF were at the same level as those of Stable Diffusion. Besides that, the WOUAF's resistance to the post-processing of images (e.g. cropping rotations, adding noise and compression) was significantly stronger than that of other methods in the state-of-the-art fingerprinting.

- Advantage: The WOUAF method is a reliable way to recognize the users of generated images while generating high-quality images and being resistant to post-processing attacks.

- Disadvantage: The method changes the generative model through weight modulation and fine-tuning of the model. Thus, it can cause an increase in computational resources need when releasing the models.

- Methodology: Using weight modulation, user-specific fingerprints are implanted into the weights of the decoder layer in Stable Diffusion. Fingerprints are generated in a mapping network, whereas a decoding network is employed to extract the fingerprint by reverse engineering from an image, which allows to determine the user who generated the image (Kim et al. 2024).

- Conclusion: The WOUAF method may be a way of incorporating user-specific fingerprinting right into the parameters of diffusion models, thereby ensuring a high degree of trust and precision in user attribution for image generation. It offers a way of validating measures to ensure the responsible use and/or regulation of generative artificial intelligence systems.

Wang et al. (2023) propose a method of training the detector that significantly improves its ability to generalize to GAN-generated images from different architectures. The problem of accurately determining if an image is real or synthetic has become majorly difficult with the continuous improvement of Generative Adversarial networks (GANs). The main challenge is that most detectors get a very high accuracy when they are tested on images produced by the

same GAN architecture as in the training, but their performance drops dramatically when these detectors are used on images generated by previously unseen GANs. This well-known problem is mainly because most detectors focus on the specific characteristics of the GANs in the training dataset and cannot generalize well.

Wang et al. (2023) propose that images made by Generative Adversarial Networks (GANs) have small, hardly noticeable "fingerprints" as artifacts from the generation process. These artifacts include operations like up-sampling. Each kind of GAN generates a distinguishing set of fingerprints that can be used to tell apart real and fake images through the use of their respective detectors. Nonetheless, since each GAN architecture will generate different kinds of fingerprints, a detector trained only on one type of GAN will most likely not be able to accurately identify images produced by a different GAN type. Thus, the authors figured that the poor performance of detectors in identifying images from a different generation architecture is due to them learning GAN-specific fingerprints instead of more generalizable patterns.

Data augmentation strategy in the fingerprint domain that changes the real fingerprints of the data to solve the problems that Wang et al. (2023) have pointed out. It is a new approach which does not rely on the traditional techniques to change the spatial properties of the pictures (e.g. cropping, blurring) but instead it changes the real fingerprints. Step one for image augmentation is to extract the original image pixels which are used in fingerprint data. This is done with a fingerprint extractor built from a convolutional autoencoder. Secondly, this fingerprint dataset of images undergoes random alterations, either via scaling or combining of two different sets, so that the newly created images can resemble the look of objects photographed by unknown/unspecified camera systems. These two steps of random perturbations of fingerprints are based on the logic that, in these ways, the set of images may be extended with the images of objects photographed by unknown/unspecified camera systems. The final batch of augmented images, including both recovered image data and their corresponding fingerprints, will be employed for training the detectors in recognizing image data as that of an object of interest.

Wang et al. (2023) published the paper describing the results of their work on improving the generalization of GAN detectors. Besides that, cross-GAN evaluation revealed that the GAN detector that was trained with augmented fingerprint images (fingerprint domain augmentation) outperformed the state-of-the-art methods in detecting fake images produced by new/unseen GANs. According to the authors, directly augmenting fingerprints in the fingerprint domain provided the GAN detector with a larger set of fake pattern distributions to learn from, which leads to improved accuracy and robustness of detection methods.

- Advantage: This novel approach helps the detector of GAN-generated images to generalize better by adding noise in the fingerprint domain, thereby enhancing the detector's performance on GAN-produced images that are not part of the training set.

- Disadvantage: Successful use of the technique is predicated on accurate fingerprint extraction using an autoencoder; any error in the extraction of fingerprints directly impacts the quality of augmented training data.

- Methodology: The proposed techniques utilize an autoencoder to extract GAN fingerprints; the extracted fingerprints are perturbed with scaling and mixup methods; the perturbed fingerprints are then combined with image reconstruction methods to create augmented fake images to be used for training a GAN detector (Wang et al., 2023).

- Conclusion: This work shows that using augmented data created from fingerprint domain information increases cross-GAN detection performance and enhances the capability of detectors to generalize to images generated by GANs that have not been seen by the detectors previously.

2.2 RESEARCH TRENDS

1. Hybrid Feature-Based Deepfake Attribution Architectures: In recent years, several studies have revealed a new trend for the use of hybrid architectures combining multiple feature types for deepfake detection and source recognition. In fact, the previous methods focused on only one feature type (either spatial artifacts or frequency characteristics) from a single domain and were not able to withstand post-processing or perform good checks against newly released generative models. Recently, there have been numerous reports about the incorporation of deep learning-based semantic and conventional forensic noise trace representations to uplift source performance and facilitate source attribution in the face of traditional method limitations. Most dual-stream architectures combine either semantic features extracted by Convolutional Neural Network (CNN) or Vision Transformer with forensic noise traces (e.g. PRNU residuals, spatial frequency artifacts). Hybrid feature representations allow the models to capture not only high-level semantic visual features, but also low-level features produced by the generative model, resulting in greatly improved source attribution performance, even after compression, noise, or other types of natural image processing manipulations.

2. Frequency-Domain and Forensic Artifact Analysis for Model Attribution: A major breakthrough is the discovery of methods for utilizing the frequency domain properties and forensic markers unique to generative models for detecting images created by such models. The use of convolutional layers, filters and upsampling techniques in Convolutional Neural Networks and Diffusion Models lead to the formation of subtle repetitive patterns in the frequency domain of their output. By carrying out transformations (i.e. performing a Discrete Cosine Transform (DCT) or a related spectral analysis), researchers are able to reveal the distinctive generative-model "fingerprints" that are left behind in the output of the model. Furthermore, the recent development of deep learning architectures and attention mechanisms have allowed researchers to better extract these features and characterize the model-specific attributes of an output by removing content-related components. This line of work elucidates how frequency-domain analysis could play a role in prevention by

substantially improving the accuracy of attribution and also providing greater flexibility to generalize over both image content and generative models.

3. Contrastive Learning and Representation-Based Detection: Current deepfake detection research mainly focuses on learning techniques to better understand the generating architectures that deepfake videos use. One of the recent trends has been to rely on contrastive and Siamese neural network based methods that use the embedding space created by models. The idea is that the learned representations for similar types of images are grouped together, while those that are quite different get separated. The latest researches into deepfakes detection are using special feature extraction methods that are tailored to real images, typically GAN-generated images and diffusion-based images. This way, they can develop classifiers that generalize well over different types of GAN-generated images and videos. These methods will allow the detection systems to not only keep up with the rapidly evolving generative AI approaches but also to develop more discriminative and transferable feature representations, rather than just depending on fixed artifact patterns.

4. Latent Feature Space Analysis Using Large Pretrained Models: Identifying synthetic images using the latent representation spaces of large pre-trained models like CLIP has recently gained considerable attention. The research community is now engaging in determining the distributions of real and generated images by using their high-dimensional embedding space, rather than simply detecting pixel-level artifacts. By being familiar with the topology and geometry of feature space, researchers through structural differences between real and synthetic images. In addition, researchers are experimenting with post-processing techniques (e.g. compression noise addition, resizing) and observe how they change the representation of an image in feature space. Besides, these results will support the creation of advanced classifiers that combine latent features and transformation-based measures to achieve generalization across different datasets and unseen generative models.

5. GAN Fingerprinting and Model-Specific Attribution Techniques: Finding and using the fingerprints of GANs, also called "deepfake forensics, " is one of the main directions in the research of deepfake and other types of AI-based image generation. Fingerprints are the

unique traces left by the architecture, training data, and initialization of particular generative models, which together form a kind of signature in the generated data. These may be learned and detected by deep learning-based classifiers so that the image can be attributed to the corresponding generative model. It has been demonstrated that such deepfake fingerprints are stable over different image domains (for example patches frequency bands, etc.) and can be used to trace an image back to its generating source even if the image is practically indistinguishable from a regular photograph.

6. Proactive Deepfake Defense Through Artificial Fingerprinting and Training Data Manipulation: Scientists are no longer satisfied with just detecting deepfakes by passively identifying them. Instead, they have started to look at ways of attributing deepfakes to their source (i.e. tracing the generated content back to its original creator). One way of doing this is by exposing parts of the generated outputs to inspectors through artificial fingerprinting or injecting something into the training data during the genesis of the generative model. These parts then become inseparable from the produced content, and hence these exposed parts can be utilized by law enforcement to locate the generated image(s) to their model of origin and/or user(s) of such a model's type. Further examples of the move towards responsible AI use in relation to generative models with embedded traceability are the invention of artificial fingerprinting, sustainable trace embedding, and user-specific weight modulation in diffusion models.

7. Robust Detection Under Real-World Image Processing Conditions: Another issue that researchers are only starting to investigate is the level of performance of detection systems in real-world environments. Often, images posted on social networks are already altered, e.g. compressed resized, or filtered before being shared. As a result, the forensic features exploited by detection systems to identify manipulated images tend to get degraded. To address this problem, researchers have built large-scale datasets which reproduce the social media processing pipeline and use them to assess detectors' ability to work under realistic conditions. This research attempts to train detectors that are resistant to such transformations. To this end, researchers are developing models that use stable generative

fingerprints and general feature representations, in contrast to relying only on delicate pixel-level artifacts.

8. Fingerprint Restoration and Cross-Model Generalization Techniques: Latest studies are also focusing on the recovery of lost generational fingerprinting features and the enhancement of cross-domain generalization. Increasingly, compression and noise are degrading the unique features used for attribution. In the first place, there are now dual domain restoration methods that not only recover the image, but also the latent fingerprint, which helps getting exact attributions even in degraded conditions. Besides, scientists are also exploring fingerprint domain data augmentation techniques to produce fingerprints from generational models that are not seen before in training so that the detectors will learn a larger set of feature distributions which results in better detection of images generated by previously unseen GAN architectures.

Chapter 3

Project

3.1 PROJECT DOMAIN

The intersection of artificial intelligence, deep learning, computer vision, and digital forensics (specifically in finding the inside of images that could have been altered using artificial intelligence) is the research project that we are doing. One of the main objectives of the research is that deep learning-based methods can be employed to study latent representations created by generative models for instance the determination of the architecture used to produce those fake (or "synthetic") images. Technological breakthroughs in generative model such as Generative Adversarial Networks and diffusion-based models have greatly enhanced the ability to produce fake images that look real at an amazing rate. The investigation is an attempt to solve the issue of identifying the source of these kinds of synthetic media by combining latent space representations and machine learning classification methods. This field of study draws from deep neural networks, generative models, and representation learning to develop systems that are more accurate, precise, and reliable at attributing the original source of deepfake media.

Helping to lay the groundwork of the future research into generative AI forensics, deepfake detection, and model attribution, this project will be highly beneficial as these areas are becoming focal points along with digital misinformation and media manipulation issues. It is a feature of these proposed technologies to tackle significantly the problem of verifying and tracing back the origins of synthetic media that is increasingly used in modern digital settings. Besides, it is planned to investigate to what extent generator-specific representations of features can be obtained and captured from latent feature signatures of GAN encoded and subsequently used for attribution purposes. Ultimately, the aim is to make use of the latent feature representations and classification of multiple generative architectures to develop even forensic tools identification of the source of AI-generated images that are scalable and efficient as well as to design a framework for accountability generation in generative AI environments.

3.2 DATASET

Synthetic facial images generated with various GAN (Generative Adversarial Network) architectures will be the source of the data for the study on deepfake attribution systems. This dataset is a mixture of images generated by each of the three main sources of generative models - ProGAN, StyleGAN2 and StyleGAN3. Typically, there are 5,000 images per each generative model which result in a balanced synthetic image dataset produced by all three generators. The goal of using such a dataset is to perform training and evaluation of models with the end result of the model being capable of identifying a specific GAN architecture that was used in creating the given synthetic image.

The dataset was created with pre-trained generative models and all three of the generators were trained on the same original data. The source data, FFHQ (Flickr-Faces-HQ) comprises high quality images of human faces that have many variations such as age, gender and everyday wear (for instance jewelry accessories etc), the kind of lighting used for the photographs, the mood/emotion shown, and finally the background in each photo. Every one of the three generative models differ in their design and the method of image generation while their training was based on FFHQ. The outcome is that each generative model generates a distinct characteristic in the images created by each machine - these distinctive fingerprints are associated with the upsampled operations, convolutional layers, as well as different training dynamics. It is the variations in upsampling, convolution, and training dynamics that help you identify which generative model was the source of a particular generated image.

This dataset possesses the proper attributes for assessing deepfake attribution systems. In fact, the synthesized images are visually indistinguishable from real human faces, but the generative network's internal mechanisms have imprinted subtle unique patterns that a machine learning model can exploit to learn discriminative features linked to each type of GAN architecture. Moreover, the dataset has images that were produced from few different models that were trained on the same source dataset, so it gives a controlled, yet very realistic and natural benchmark to do a study on the GAN's fingerprints and deepfake attribution of models that will help the attribution method developer to evaluate how well they can distinguish GANs

3.3 GENERATIVE ADVERSARIAL NETWORKS (GANs)

Generative Adversarial Networks (GANs) are a type of deep learning models that produce synthetic. Using a "twisting" mechanism to the real data are the basis of the synthetic data creation by the generative adversarial networks (also known by their acronym GANs). Initially, this concept was presented by Ian Goodfellow in the year 2014 and it involves 2 neural networks, interacting antagonistically in a training process named a minimax or zero-sum game. In that game, the first neural network (that acts as the generator) produces synthetic samples using specific inputs of random numbers (noise). Second neural network (the discriminator) will get also these inputs and it will say whether they are real or generated. Due to this adversarial mechanism, the generator's skill in producing credible synthetic outputs grows very fast; as such, GANs are among the most powerful categories of generative models in contemporary artificial intelligence applications.

The generator part of a typical GAN (Generative Adversarial Network) works by taking in a layer of random noise as input. It then performs a number of neural computations (layers) and finally outputs a synthetic image. On the other hand, the GAN's discriminator takes in both genuine images from the dataset and synthetic images made by the generator, and tries to label them as either real or fake. While the generator tries to trick the discriminator with the most realistic generated image possible, the discriminator, in turn, gets better and better at telling the two apart. Through this learning from each other process, the generator gets to know the training data distribution and the result of the generator's photos is on a par with the original data quality.

The training of GANs is performed using a minimax optimization objective. The generator (G) and the discriminator (D) are attempting to respectively maximize or minimize their probability of being correct. That is, the discriminator is attempting to maximize its probability of accurately identifying real and generated (i.e., fake) samples while G is trying to minimize the chances of being accurately identified by D as either real or generated sample. This minimax optimization problem can be represented mathematically as:

$$\min_{G}\max_{D} V(D,G) = E_{x\sim pdata(x)}[logD(x)] + E_{z\sim pz(z)}\left[\log\left(1 - D\left(G(z)\right)\right)\right]$$

where:

- x: A real image sampled from the training dataset.
- z: A random latent vector sampled from a prior distribution (typically Gaussian or uniform).
- G(z): The synthetic image generated by the generator from latent vector z.
- $D(x)$: The probability that the discriminator assigns to an image being real.
- $p_{data}(x)$: The distribution of real images in the training dataset.
- $p_z(z)$: The probability distribution used to sample latent vectors.

Both the generator and discriminator improve their ability to produce and recognize accurately the realistic images when trained in an iterative process. Eventually, the generator's ability to produce a photorealistic image exits the limits of human perception.

One of the key points of Generative Adversarial Networks (GANs) is their wide applicability. For instance, they can be used for creating new images, transferring the style from one image to another, increasing the resolution of images, or even making deepfakes. There are many versions of GANs which have been improved and developed like ProGAN, StyleGAN2, and StyleGAN3 that can produce images of a high quality and very realistic in appearance, something that was not possible before using a GAN. Nevertheless, besides the high quality and realism, the generative architecture that is responsible for producing these synthetic images reflects small identifiable traces on the images which have been generated. Such identifying traces may, for example, include structural differences in the GAN architecture, training duration, and upsampling methods. This research makes use of identifying the traces left by the GAN in generating each synthetic image in order to identify the latent representation of the synthetic images; hence, enabling the attribution of deepfake generative architectures which can be helpful in the digital forensics processes for the analysis of digital media.

3.4 HARDWARE AND SOFTWARE REQUIREMENTS

A blended compute implementation of this project, for example, can be a cloud-based Kaggle Notebook environment paired with a local workstation. Indeed, a local machine was predominantly used for creating synthetic images with pre-trained GANs whereas the Kaggle environment was mainly used in executing the computationally demanding functions of latent feature extraction and training classifiers. Kaggle provides a GPU-enabled cloud-based computing environment which alleviates the computational load on local hardware; thus, experiments with machine learning models have become much more efficient.

Both a local and cloud-based environment in tandem made it possible for the project to weigh computational efficiency and available resources. The work with pre-trained generative models for image generation (for instance ProGAN StyleGAN2, StyleGAN3) was done locally while the Kaggle environment was used for feature extraction and training of machine learning classifiers. In this way, the project was able to take advantage of local computer power and at the same time of cloud-based GPU resources for quick deep-learning experimentation.

3.4.1. HARDWARE REQUIREMENTS

For this work, a mix of Kaggle's cloud GPU resources and accessible local machine hardware was used. The local hardware was employed for producing synthetic images from pretrained GAN models. To produce synthetic images, sufficient memory and processing power were required not only to run a deep learning framework but also to handle the volume of real data that could be used for generating synthetic images. The Kaggle Notebook environment was used for extracting latent signatures and assembling classification models to identify whether a generated image was created by a GAN or was from some other source. Below is a list of typical hardware resources used throughout the project:

- CPU Processing Unit (CPU): Multi-Core Processor with at least 6 cores.
- Graphics Processing Unit (GPU):
    - Kaggle GPU environment (e.g., NVIDIA Tesla P100 / T4) used for model training and feature extraction
    - Graphics Card with atleast 4GB VRAM
- Memory (RAM): 16GB RAM
- Storage: 200GB

The hybrid use of local hardware for image generation and cloud GPU resources for model training enabled efficient experimentation while reducing the need for expensive dedicated hardware.

3.4.2. SOFTWARE REQUIREMENTS

The software environment of this project consists of several machine learning, deep learning, and data processing tools. The deepfake attribution pipeline was developed primarily using Python-based frameworks in a Kaggle Notebook environment that provides preinstalled libraries and the possibility to use GPU to speed up deep learning operations. The main software components used in this project are:

- Operating System:
    - Kaggle cloud environment running Linux-based systems
    - Local workstation environment for running GAN image generation scripts
- Programming Language:
    - Python 3.8 and Python 3.9
- Deep Learning Framework:
    - PyTorch for implementing neural networks and feature extraction pipelines
    - Pretrained GAN implementations including ProGAN, StyleGAN2, and StyleGAN3
    -

- Machine Learning Libraries:
    - Scikit-learn for training classification models and evaluating performance
- Data Processing Libraries:
    - NumPy for numerical computations
    - Pandas for dataset handling and preprocessing
- Image Processing Libraries:
    - OpenCV and Pillow for image loading and preprocessing
- Notebook Environment:
    - Kaggle Notebook (Jupyter-based environment) for running experiments, training models, and analyzing results

This software setup allowed the project to utilize GPU acceleration through Kaggle while maintaining a flexible Python-based development environment for implementing deep learning and machine learning workflows.

3.5 METHODOLOGY

The proposed methodology adopts a multi-stage framework to provide an innovative approach for classifying visual assets as being created by fake sources and such classification will be done through the analysis of latent features of the asset. The aim is to identify which generative architecture is responsible for generating a particular synthetic image. The system is planned to have modules for detection of a persons face; detection of deepfake images and creation of a database of facial images to compare a latent feature and classify images of facial features. By fully processing the images, which involves extracting the facial region of interest from the image, deepfake detection, and then latent feature extraction that identifies the generative model that created the deepfake, we offer a complete solution for deepfake images. The details of each of the system components are provided below.

3.5.1. MODULE DEVELOPMENT AND INTEGRATION

1. Face Detection Module

The facial area is identified and extracted from the input images during the initial stage of the process. As the goal of this work is to trace deepfake to face, it is imperative that the facial area is isolated from the raw input image first. Then RetinaFace model was used to detect facial region in the next step. Facial recognition with a single step detection algorithm such as the one offered by RetinaFace is highly accurate and robust even under challenging conditions hardly seen in real world such as head orientations, different lighting conditions and partial occlusions. RetinaFace gives the coordinates of the bounding box that encloses the face. Besides that, it points out several facial landmarks such as eyes, nose, and mouth. After face has been accurately located by the model, it will be cropped and resized to a uniform resolution which is good for the next steps. This preprocessing stage permits the models downstream to operate exclusively on facial features, hence increasing both their efficiency and accuracy in detecting as well as attributing deepfakes.

2. Deepfake Detection Module

One cannot simply upload an image to the system and receive the creator information. It's a stepwise process where you first need to do a check on the image itself. Is the image

uploaded, a real one or a synthetic one (artificially created)? Only images identified as deepfake images should be processed by the attribution model. In place of our custom detection model, we chose to use a pretrained deepfake detection model called UniversalFakeDetect that was first unveiled at CVPR 2023. UniversalFakeDetect is designed to be universal; it runs across many generative models and can tell synthetic images made by different GAN and diffusion-based architectures apart. The cropped facial images obtained from the first face detection phase will be sent to the UniversalFakeDetect model to decide whether to label the input image as real or fake. Every image that is labeled as real at this stage will be released while every image that is labeled as synthetic at this stage will be forwarded to the attribution module to find out who the creator of these images is. Therefore, the attribution module serves as a filter that the attribution system will work with only those images for which the confirmation of the artificial generative model origin has been obtained.

3. Synthetic Image Dataset Generation Module

The following step is to generate a dataset composed only of artificial images that will serve to educate the deepfake attribution system.The production of the face images is going to be done with the help of the following three GAN architectures and their pretrained generators:

- ProGAN (Progressive Growing GAN)
- StyleGAN2
- StyleGAN3

These three models represent different generations of GAN architectures and were used to produce synthetic images with very distinct characteristics. All generators were originally pretrained to produce images using the Flickr-Faces-HQ(FFHQ) dataset, which is composed of high-quality face images and many different types of attributes (age, sex, lighting, expression, background, etc.). Approximately 5,000 synthetic images were generated per generator, giving a balanced set of images to train on, with images labelled for the particular generator used to create them. This dataset provides information on the

subtle variations generated by each generator during the image creation, which will help the model learn how to identify the patterns created between the various generative architectures.

4. Latent Signature Extraction and Feature Preprocessing Module

This module intends to derive compact numerical representations called latent signatures from deepfake images which will disclose the unique elements of the generator at each step of the deepfake image creation process. Rather than employing pixel values of high-dimensional images for the image classification task, latent representations of images will be produced, changed into structured feature vectors, and will be apt for ML models. The latent representation of an image will be produced using the encoder4editing (e4e) system which is a GAN inversion model specially created for projecting images into the latent space of StyleGAN-based generators. The e4e encoder model was pretrained on the FFHQ dataset and it is capable of efficiently projecting facial images into the W+ latent space of StyleGANs.

Prior to performing latent extraction, the raw images need to go through preprocessing. Every raw image has to first be resized to 256 x 256 pixels, then transformed into tensor form, and lastly, the pixel values must be normalized by applying the following operations:

- Resize image to 256 × 256 resolution
- Convert image to tensor representation
- Normalize pixel values using mean = 0.5 and standard deviation = 0.5 for all RGB channels

Normalizing the raw images makes the raw images compatible with the trained encoding model used for latent feature extraction. Besides, through normalization, the behavior of extracting the latent features from the raw images becomes stable.

After the preprocessing is done, each image goes through an encoder network named e4e.he W+ latent space contains multiple latent representations for each generator layer.

For instance, in StyleGAN architecture, the latent encoding typically consists of eighteen layers with a 512-dimensional vector for each layer. To keep the feature representation compact and consistent across different GANs, the extraction process gets the latent statistical features of the first fourteen layers of the W+ latent code. There are five different statistical descriptors that are computed for each layer:

- Mean of the latent vector
- Standard deviation of the latent vector
- L2 norm of the latent vector
- Minimum value of the latent vector
- Maximum value of the latent vector

Together, these five descriptors will encompass the general distribution of latent values of the generator's layers along with their structural characteristics. Since the statistics are calculated on 14 layers, the resulting feature vector will be of the following size:

14 layers × 5 statistical features = 70-dimensional latent signature

As a result, the final 70-dimensional vector will act as the latent fingerprint of the generated images. The extraction process will be conducted for all images generated by the three different GAN architectures used in the dataset.

The specified feature labels depend on the kind of architecture that generated the feature under investigation. This effort will yield a set of features extracted in a record-type CSV file containing 70 latent signature features and one label indicating the architecture used to generate the image. The final feature set will be organized as follows:

(f0, f1, f2, ..., f69, label)

Where:

- f0 – f69 represent the extracted latent signature features
- label indicates the GAN architecture responsible for generating the image

The compiled feature dataset is the input to GAN attribution classification systems. Instead of using the raw image itself, by relying on the statistical signatures that correspond to the latent-feature signature of the input image, the model can maintain and express the unique characteristics of an individual generator in latent space, thereby enabling it to differentiate accurately between images created by different GAN architectures.

5. GAN Attribution Classification Module

The machine learning models are trained at the final stage of the system to classify the latent attributes of the images produced by different Generative models (GANs) according to their type-based generative architecture; thus, providing a mapping (association) of each generated image to the generative model that generated it. Each generative model has been trained at the previous level, and provides the classifier with the latent feature representation of its generated image as the input feature for determining how to classify successive images generated by other Generative models using the feature representations in this stage. The Kaggle notebook environment has been employed for developing an end-to-end / full life cycle Classifier, utilizing Python, the PyTorch framework, and all associated machine learning libraries provided by Scikit-learn. The entire process includes steps such as feature extraction, dataset development/creation preprocessing model training, and model evaluation/validation.

The first step in preparing a feature dataset involved scanning through the directories delivered with images created by three GAN architectures: ProGAN, StyleGAN2, and StyleGAN3 - here, every subfolder is a class label. For each picture, a latent signature extraction method takes the latent input and gives back a one dimensional latent feature vector which comes from the output of an encoder. Such vectors are kept in a feature matrix X, while the labels representing GAN architectures are kept in y, so the classifier can find patterns specific to each architecture. The second step consists of a resolution-

independent augmentation strategy which is intended to keep the deep model from being too reliant on fixed image resolutions - it is experimentally shown that such a method leads to improving the overall system robustness. Each image is first randomly resampled at a resolution level in the range 64 64, 1024 1024, then re-resampled a second time at the level 256 256; after that, the individual is converted to a tensor and normalized to the values of mean [0.5, 0.5, 0.5] and standard deviation [0.5, 0.5, 0.5]. Such preparation is done for images existing in nature and thus the latent extraction model aims to be immune to changes in resolution.

First, the feature dataset will be split into two parts - one for training and one for testing the attribution model's performance. Also, the splitting will be done using stratified sampling so that each GAN architecture - per class - has a statistically similar number of samples.

- Training set: 80% of the data
- Testing set: 20% of the data
- Random seed: 42 (for reproducibility)
- Stratification: Maintains balanced representation of each GAN class

Feature scaling will be done to each Latent Vector before training each classifier. The standard procedure for Feature Scaling is "Standardization", which will be done using the Scikit-learn package. The equation for the transformation is given below

$$X' = \frac{X - \mu}{\sigma}$$

Where:

- μ represents the mean of each feature
- σ represents the standard deviation of each feature

Standardizing the features enables all the Latent Features to be on the same scale, which leads to better machine learning model performance and stability (e.g. less training and testing times) - with Logistic Regression and Support Vector Machines.

Various machine learning classifiers were trained and compared to each other to find out the extent to which latent signatures can precisely identify the type of Generative Adversarial Network (GAN) used to generate each of the 7 datasets. The classifiers, that were utilized for the experimental purposes, are shown as follows:

1. Logistic Regression

Logistic Regression is a form of supervised learning in machine learning that is primarily used for the purpose of classification. It estimates that probability that a given feature vector is a representative of a member of a specific class by using the logistic (sigmoid) function. At the present project, Logistic Regression is applied to model the association between latent feature vector and the label assigned to a specific GAN architecture. Structure-wise and efficiency-wise, Logistic Regression model offers a good baseline for assessing if latent signatures are distinct enough to be separated.

2. Support Vector Machine (SVM)

Support Vector Machine is a kind of supervised classification algorithm that separates the data points by the best possible decision area of classification (or the class boundaries), known as the hyperplane. What an SVM model aims for is to separate the classes with the greatest distance or margin between the classes. First, linear SVM models were trained with the latent signature feature vectors of the GAN images, so the SVM model would be able to notice the differences between the images produced with different GAN architectures.

3. Random Forest

Random Forest is an ensemble method; as such, it creates many decision trees and uses the outputs from all of these different trees to develop a single prediction about the data input. For this reason, each tree can be trained on a randomly created subset of the data

and thereby create a more robust model while reducing the incidence of overfitting. This project will use the Random Forest Classifier (RFC) to capture nonlinear relationships and assist in improving classification performance regarding predicting the source of images generated by GANs (Generating Adversarial Networks).

Each of the Individual Trained Classifiers will undergo evaluation after being applied to the Test Data using the following performance metrics:

- Classification Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

The confusion matrix is a method used to visually represent the model's classification accuracy of images generated by different architectures of GANs. The Classifier having the highest accuracy will be selected as the best-performing attribution model.

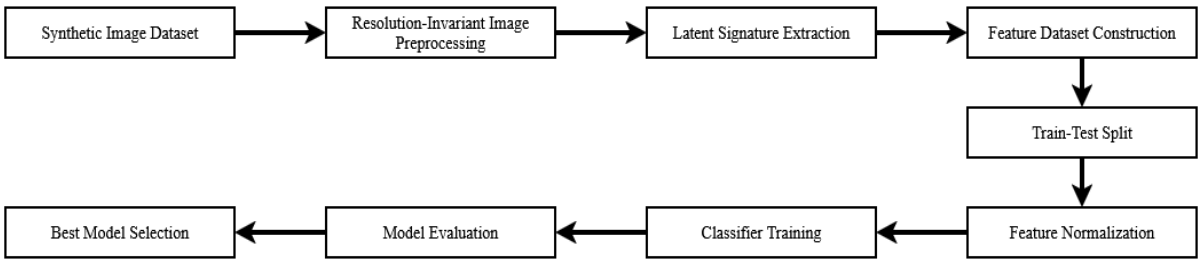


Figure 1: Methodology Pipeline

3.5.2. ARCHITECTURE DIAGRAM

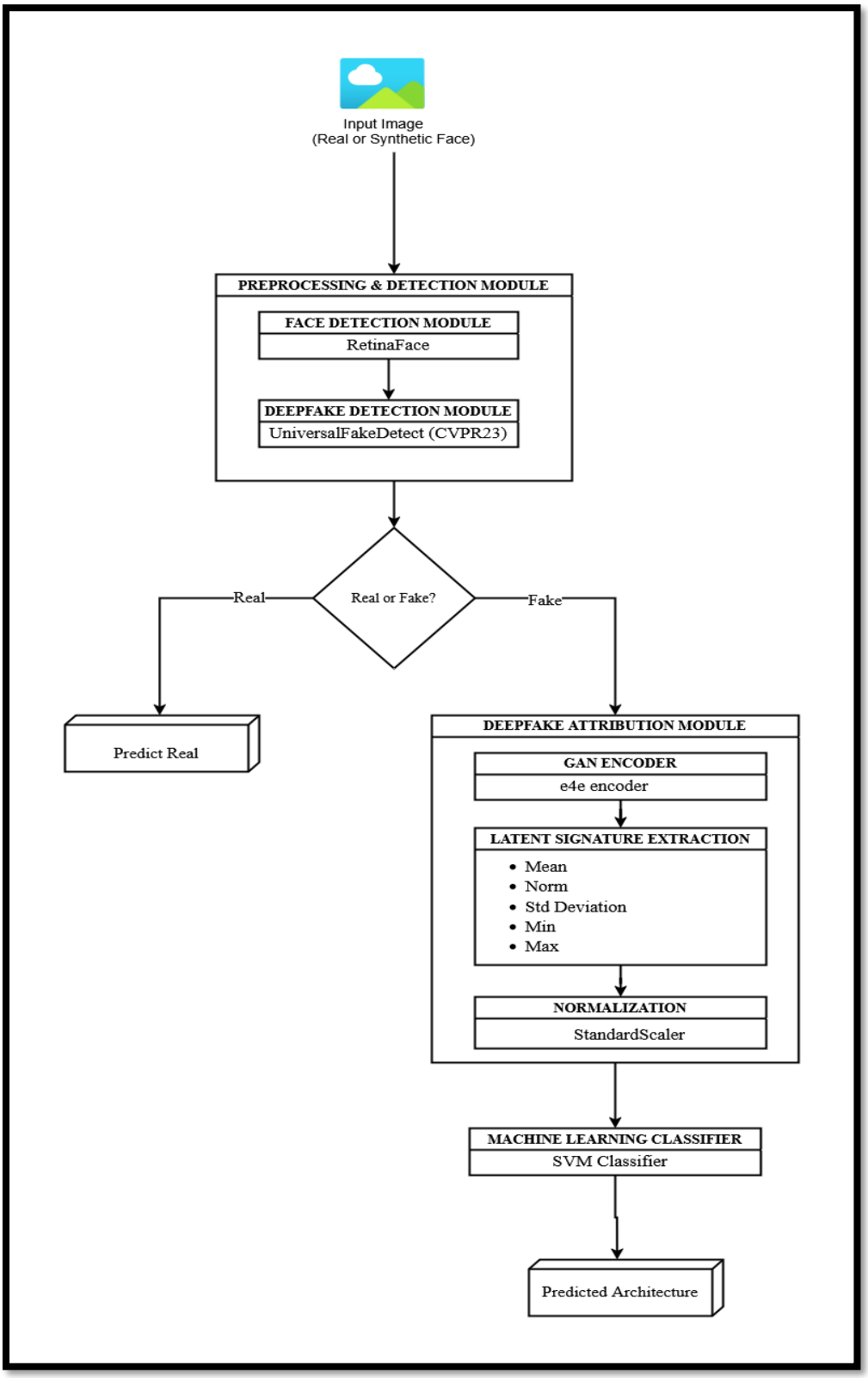


Figure 2: Architecture Diagram

3.5.3. COMPARISON AND RESULTS

Three classification models of latent components have been put head to head: Logistic Regression, Random Forest Classifier, and Support Vector Classifier (SVC). Here are the comparison results of every model:

1. Logistic Regression Results:

Logistic regression is the model that will be used as a baseline in the GAN attribution task. To make sure that the model will actually be a proper baseline, it was given the opportunity to perform up to 3000 training iterations in order to find the most optimum point of convergence. Here is a recap of the performance:

class	precision	recall	f1-score	support
progan	0.99	0.98	0.99	1000
stylegan2	0.99	1.00	0.99	1000
stylegan3	0.99	0.99	0.99	1000

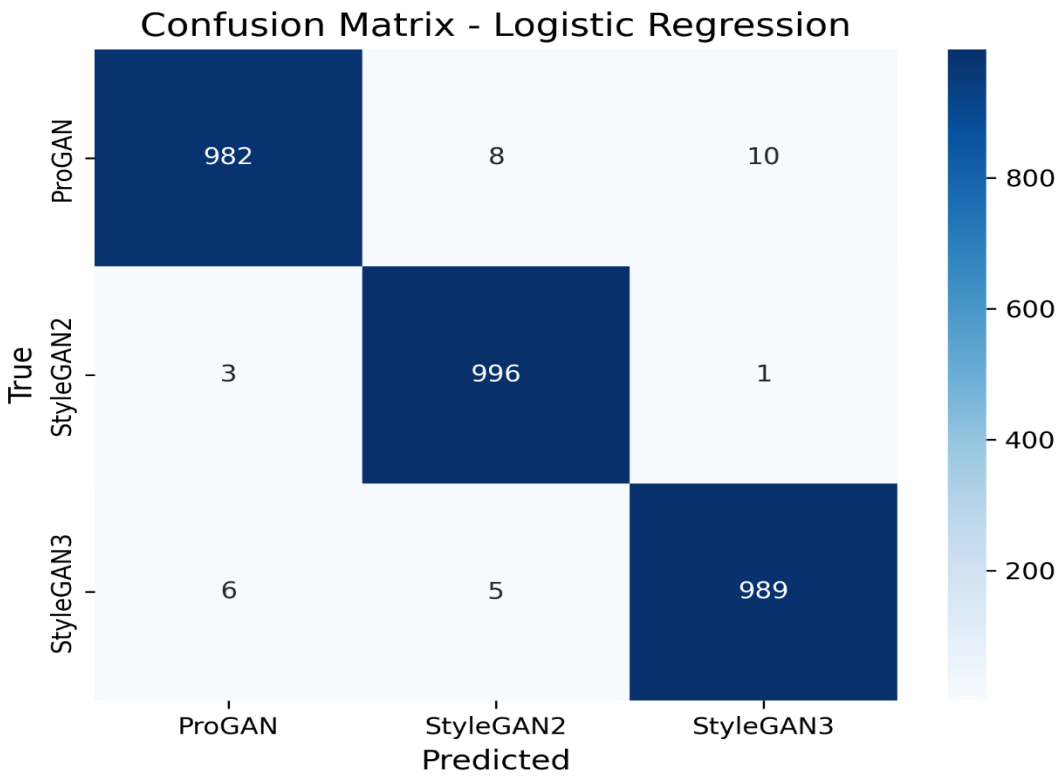


Figure 3: Linear Regression Confusion Matrix

- Accuracy: 98%
- Performance is strong due to the linear separability of latent features
- Some confusion occurs between StyleGAN2 and StyleGAN3, which share similar architecture characteristics.

Logistic Regression demonstrates that the extracted latent signatures already contain significant information for distinguishing between GAN architectures. Further reference to classification report may be found in appendix 4.

2. Support Vector Machine (SVM) Results:

A linear SVM classifier was trained using balanced class weights to improve generalization across GAN classes. The performance summary is as follows:

class	precision	recall	f1-score	support
progan	1.00	0.98	0.99	1000
stylegan2	0.99	1.00	0.99	1000
stylegan3	0.99	0.99	0.99	1000

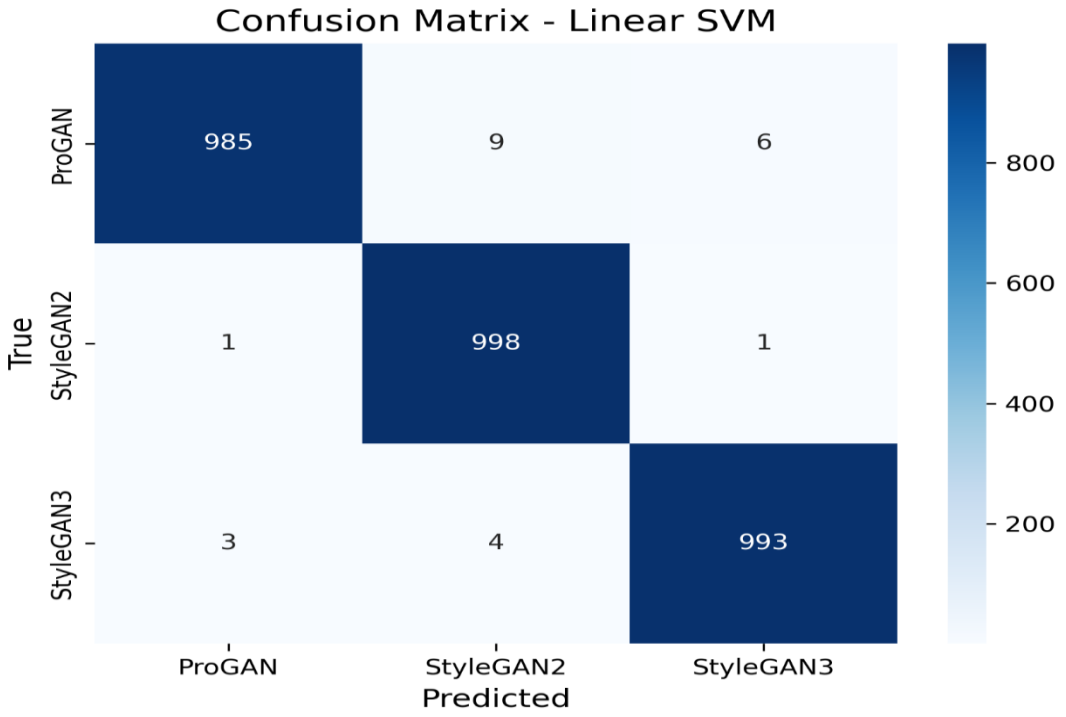


Figure 4: Linear SVM Confusion Matrix

- Accuracy: 99%
- Highest performance among the tested models
- Very effective in high-dimensional feature spaces

SVM performs well because it maximizes the decision boundary margin between latent feature clusters belonging to different GAN architectures. Further reference to classification report may be found in appendix 5.

3. Random Forest Results:

The Random Forest classifier is capable of detecting different nonlinear components of feature space and it is composed of 200 decision trees that are intertwined. Random

Forest classification results are as follows:

	precision	recall	f1-score	support
progan	0.91	0.93	0.92	1000
stylegan2	0.94	0.93	0.93	1000
stylegan3	0.95	0.94	0.95	1000

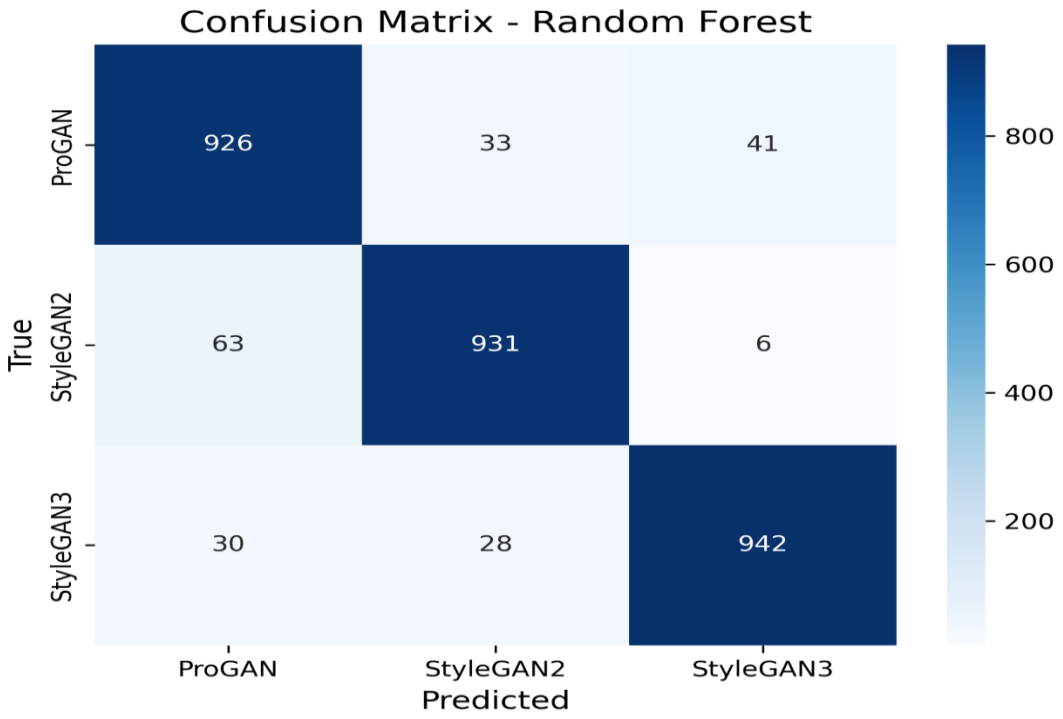


Figure 5: Random Forest Confusion Matrix

- Accuracy: 93%
- Slightly lower than SVM but still strong
- Handles feature interactions effectively

The Random Forest classifier is resilient to noise and provides stable classification performance across the all of the data. For the complete classification report, please see Appendix 6.

Among the tested models, the SVM was found to have the highest overall accuracy and performance; therefore, it will be used as the final classifier for the GAN attribution model.

Chapter 4

Conclusion & Future Work

4.1. CONCLUSION

This project researched on deepfake model attribution, i.e. the task of identifying which generative architecture (i.e. how the synthetic image was generated) created a particular synthetic image. We have four major components in the system: This project will mainly contribute that we have utilized the latent space representation of GAN-generated images to generate an architecture-specific fingerprint.

ProGAN, StyleGAN2, and StyleGAN3 were the source of the synthetic datasets. Statistical features hidden in latent spaces of these datasets were extracted by Encoders, and classifiers Logistic Regression, Support Vector Machine (SVM), Random Forest model were trained on them to differentiate between GAN architectures. Results of the experiments demonstrated that each classifier had its own level of success in classifying latent signatures; however, among the three, the SVM classifiers outperformed the other two at pinpointing differences in GAN architecture.

The report states that latent representations generated by generative models differ from each other by small structural variations and thus could be used for attribution. Also, the project has shown that it is challenging to differentiate between very similar architectures (e.g. StyleGAN2 versus StyleGAN3). Overall, the directorate has proved that integrating latent feature analysis and machine learning classification can be very effective techniques for attributing deepfakes in digital media forensics.

4.2. FUTURE WORKS

The current project has demonstrated the potential of using latent signatures as a method to break down generative adversarial networks (GANs), but there are still many areas in which it could be expanded and improved at the same time.

- Including more generative models:
  The current implementation focuses on three GAN architectures: ProGAN, StyleGAN2, and StyleGAN3. Future work could include additional generative models such as StyleGAN-XL, diffusion models (Stable Diffusion), or other GAN variants. Expanding the range of architectures would allow the system to be evaluated in more realistic scenarios where many generative models exist.

- Improving cross-architecture generalization:
  Although the system successfully differentiates between the selected GAN models, some confusion was observed between architectures with similar design principles. Future work could explore advanced feature extraction methods or deep neural classifiers to improve the system's ability to distinguish between closely related generative architectures.

- Robustness to real-world image transformations:
  A lot of changes take place regarding photographs or other images that were taken y the camera in an ordinary manner, these changes include changes to how the image is stored, the size of the photograph, and the addition of unwanted noise. There are also groupings in the image to include social networking applications. In order to evaluate future improvements for the robustness of image attribution for photographs (Through data augmentation or for methods of training) under conditions where image transformation will occur.

- Larger and more diverse datasets:
  The existing datasets consist of face images that were created using pretrained models. In the future, research can look into creating larger and more diverse datasets that not

only have face images produced in different training ways but also based on different prompts so that the scalability and generalizability of the attribution systems can be assessed.

- Integration with real-time forensic systems:
  Our current efforts are concentrated on running offline experiments of image attribution. One of the directions for future research could be the integration of the attribution framework which is suggested here into the real-time digital forensic systems in order to detect and attribute synthetic media that are being shared on social media and other online platforms.

By pursuing research in the above directions, future studies will improve the ability of reliable and scalable deepfake attribution systems to keep up with the proliferation of generative AI in our daily lives.

REFERENCES

[1]. Chopade, A., Vhasale, P., Waghmare, M. and Belgamwar, P. (2025). A pattern-based comparative analysis and hybrid model for deepfake image source attribution. Proc. 6th International Conference on Intelligent Communication Technologies and Virtual Mobile Networks (ICICV), IEEE, pp. 557–563.

[2]. Zhang, J., Wang, Y., Tohidypour, H.R. and Nasiopoulos, P. (2022). Deep learning-based synthesized image attribution using frequency distribution information. Proc. International Conference on Computational Science and Computational Intelligence (CSCI), IEEE, pp. 1566–1567.

[3]. Pontorno, O., Guarnera, L., and Battiato, S. (2025). DeepFeatureX-SN: Generalization of deepfake detection via contrastive learning. Multimedia Tools and Applications, Springer, pp. 1–20.

[4]. Uhlenbrock, L., Bergmann, S., and Riess, C. (2025, September). Latent Landscapes: Topology of the CLIP Feature Space for Synthetic Image Detection. In Proceedings of the 33rd European Signal Processing Conference (EUSIPCO), IEEE, pp. 51–55.

[5]. Zhu, L., Ye, K., Yao, J., Li, C., Zhao, L., Cao, Y., Wang, D., and Hao, J. (2025, April). MAID: Model Attribution via Inverse Diffusion. In Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), IEEE, pp. 1–5.

[6]. Guarnera, L., Giudice, O., Nießner, M., and Battiato, S. (2022). On the exploitation of deepfake model recognition. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW), pp. 61–70.

[7]. Dell'Anna, S., Montibeller, A., and Boato, G. (2025, June). TrueFake: A real world case dataset of last generation fake images also shared on social networks. In 2025 International Joint Conference on Neural Networks (IJCNN), IEEE, pp. 1–8.

[8]. Fan, C., Shen, A., Han, Z., Tong, C., Wang, Z., and Yi, D. (2024, July). Dual-domain multi-model GAN fingerprint restoration for compressed fake face attribution. In Proceedings of the IEEE International Conference on Multimedia and Expo (ICME), IEEE, pp. 1–6.

[9]. Yu, N., Davis, L. S., and Fritz, M. (2019). Attributing fake images to GANs: Learning and analyzing GAN fingerprints. In Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), pp. 7556–7566.

[10]. Hirofumi, S., Fukuchi, K., Akimoto, Y., and Sakuma, J. (2022, July). Did you use my GAN to generate fake? Post-hoc attribution of GAN generated images via latent recovery. In 2022 International Joint Conference on Neural Networks (IJCNN), IEEE, pp. 1–8.

[11]. Tariang, D., Corvi, R., Cozzolino, D., Poggi, G., Nagano, K., and Verdoliva, L. (2024). Synthetic image verification in the era of generative artificial intelligence: What works and what isn't there yet. IEEE Security & Privacy, Vol. 22, No. 3, pp. 37–49.

[12]. Yu, N., Skripniuk, V., Abdelnabi, S., and Fritz, M. (2021). Artificial fingerprinting for generative models: Rooting deepfake attribution in training data. In Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), pp. 14448–14457.

[13]. Sun, P., Li, Y., Qi, H., and Lyu, S. (2022, October). FakeTracer: Exposing deepfakes with training data contamination. In 2022 IEEE International Conference on Image Processing (ICIP), pp. 1161–1165. IEEE.

[14]. Kim, C., Min, K., Patel, M., Cheng, S., and Yang, Y. (2024). WOUAF: Weight modulation for user attribution and fingerprinting in text-to-image diffusion models. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 8974–8983.

[15]. Wang, H., Fei, J., Dai, Y., Leng, L., & Xia, Z. (2023, July). *General GAN-generated image detection by data augmentation in fingerprint domain*. In 2023 IEEE International Conference on Multimedia and Expo (ICME) (pp. 1187–1192). IEEE.

Appendices

APPENDIX 1: Latent Signature Extraction Code

```
def extract_latent_features(image_path):

 img = Image.open(image_path).convert("RGB")
 img = transform(img).unsqueeze(0).to(device)

 with torch.no_grad():
        latent = encoder(img)

 latent = latent.cpu().numpy()

 features = []

 # Use first 14 latent layers
 for layer in latent[0][:14]:
        features.extend([
                layer.mean(),
                layer.std(),
                (layer**2).sum()**0.5,
                layer.min(),
                layer.max()
        ])

 return features
```

This function projects the input image into the W+ latent space and computes statistical features across latent layers.

APPENDIX 2: Dataset Feature Generation

```
import os

X = []
y = []

for label in ["ProGAN", "StyleGAN2", "StyleGAN3"]:
    folder = os.path.join(dataset_path, label)

    for img_name in os.listdir(folder):
        img_path = os.path.join(folder, img_name)

        features = extract_latent_features(img_path)

        X.append(features)
        y.append(label)
```

This process converts the image dataset into latent feature vectors used for classifier training.

APPENDIX 3: Image Preprocessing Pipeline

```
transform = transforms.Compose([
    transforms.Resize((256, 256)),
    transforms.ToTensor(),
    transforms.Normalize([0.5, 0.5, 0.5],
                         [0.5, 0.5, 0.5])
])
```

This preprocessing step ensures the input image format matches the encoder training conditions.

APPENDIX 4: Logistic Regression Classification Report

	precision	recall	f1-score	support
progan	0.99	0.98	0.99	1000
stylegan2	0.99	1.00	0.99	1000
stylegan3	0.99	0.99	0.99	1000
accuracy			0.99	3000
macro avg	0.99	0.99	0.99	3000
weighted avg	0.99	0.99	0.99	3000

APPENDIX 5: Linear SVM Classification Report

	precision	recall	f1-score	support
progan	1.00	0.98	0.99	1000
stylegan2	0.99	1.00	0.99	1000
stylegan3	0.99	0.99	0.99	1000
accuracy			0.99	3000
macro avg	0.99	0.99	0.99	3000
weighted avg	0.99	0.99	0.99	3000

APPENDIX 6: Random Forest Classification Report

	precision	recall	f1-score	support
progan	0.91	0.93	0.92	1000
stylegan2	0.94	0.93	0.93	1000
stylegan3	0.95	0.94	0.95	1000
accuracy			0.93	3000
macro avg	0.93	0.93	0.93	3000
weighted avg	0.93	0.93	0.93	3000

APPENDIX 7: Hyperparameter Settings

Model	Parameters
Logistic Regression	max_iter=3000
SVM	kernel=linear, class_weight=balanced
Random Forest	n_estimators=200